Integration Part 3 Further Integration - Homework Answers

Revision | Warm-up

Each inverse trigonometric value must lie in the range of the function: $-\frac{\pi}{2} \leq \sin^{-1} x \leq \frac{\pi}{2}$, $0 \leq \cos^{-1} x \leq \pi$, $-\frac{\pi}{2} < \tan^{-1} x < \frac{\pi}{2}$.

1 Exercise 17B Q2

b $\sin^{-1} \frac{1}{2} = \frac{\pi}{6}$, because $\sin \frac{\pi}{6} = \frac{1}{2}$ and $-\frac{\pi}{2} \leq \frac{\pi}{6} \leq \frac{\pi}{2}$.

d $\arctan 1 = \frac{\pi}{4}$, because $\tan \frac{\pi}{4} = 1$ and $-\frac{\pi}{2} < \frac{\pi}{4} < \frac{\pi}{2}$.

e $\sin^{-1}(-1) = -\frac{\pi}{2}$, because $\sin\left(-\frac{\pi}{2}\right) = -1$ and $-\frac{\pi}{2}$ is in the range.

f $\cos^{-1} 0 = \frac{\pi}{2}$, because $\cos \frac{\pi}{2} = 0$ and $0 \leq \frac{\pi}{2} \leq \pi$.

i Let $\alpha = \sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$. Then $\sin \alpha = -\frac{\sqrt{3}}{2}$, where $-\frac{\pi}{2} \leq \alpha \leq \frac{\pi}{2}$. So α is in the fourth quadrant with related angle $\frac{\pi}{3}$, and $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) = -\frac{\pi}{3}$.

j Let $\alpha = \cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)$. Then $\cos \alpha = -\frac{1}{\sqrt{2}}$, where $0 \leq \alpha \leq \pi$. So α is in the second quadrant with related angle $\frac{\pi}{4}$, and $\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right) = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$.

k Let $\alpha = \arctan\left(-\frac{1}{\sqrt{3}}\right)$. Then $\tan \alpha = -\frac{1}{\sqrt{3}}$, where $-\frac{\pi}{2} < \alpha < \frac{\pi}{2}$. So α is in the fourth quadrant with related angle $\frac{\pi}{6}$, and $\arctan\left(-\frac{1}{\sqrt{3}}\right) = -\frac{\pi}{6}$.

l $\arccos(-1) = \pi$, because $\cos \pi = -1$ and $0 \leq \pi \leq \pi$.

2 Exercise 17B Q5

a $\sin \frac{4\pi}{3} = -\frac{\sqrt{3}}{2}$ (third quadrant, related angle $\frac{\pi}{3}$), so $\sin^{-1}\left(\sin \frac{4\pi}{3}\right) = \sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) = -\frac{\pi}{3}$. The answer is not $\frac{4\pi}{3}$, which is outside the range of $\sin^{-1}$.

b $\cos\left(-\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$, so $\cos^{-1}\left(\cos\left(-\frac{\pi}{4}\right)\right) = \cos^{-1}\frac{1}{\sqrt{2}} = \frac{\pi}{4}$.

c $\tan \frac{5\pi}{6} = -\frac{1}{\sqrt{3}}$ (second quadrant, related angle $\frac{\pi}{6}$), so $\tan^{-1}\left(\tan \frac{5\pi}{6}\right) = \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) = -\frac{\pi}{6}$.

d $\cos \frac{5\pi}{4} = -\frac{1}{\sqrt{2}}$ (third quadrant, related angle $\frac{\pi}{4}$), so $\cos^{-1}\left(\cos \frac{5\pi}{4}\right) = \cos^{-1}\left(-\frac{1}{\sqrt{2}}\right) = \frac{3\pi}{4}$.

e $2\sin\left(-\frac{\pi}{6}\right) = 2 \times \left(-\frac{1}{2}\right) = -1$, so $\arcsin\left(2\sin\left(-\frac{\pi}{6}\right)\right) = \arcsin(-1) = -\frac{\pi}{2}$.

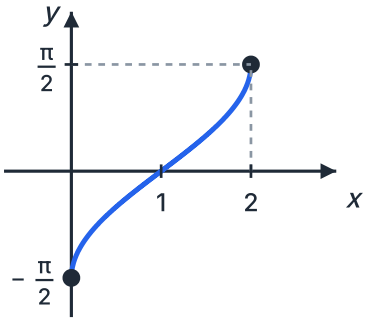
f $3\tan \frac{7\pi}{6} = 3 \times \frac{1}{\sqrt{3}} = \sqrt{3}$ (third quadrant, where $\tan$ is positive), so $\arctan\left(3\tan \frac{7\pi}{6}\right) = \arctan\sqrt{3} = \frac{\pi}{3}$.

3 Exercise 17C Q2

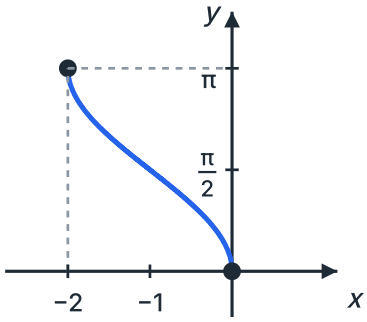
a $y = \sin^{-1}(x-1)$ is $y = \sin^{-1} x$ shifted right 1 unit. It needs $-1 \leq x-1 \leq 1$, so the domain is $0 \leq x \leq 2$. The range is $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$. Key points: $\left(0, -\frac{\pi}{2}\right)$, $(1, 0)$, $\left(2, \frac{\pi}{2}\right)$. The function is neither even nor odd, because its domain is not symmetric about the y-axis.

b $y = \cos^{-1}(x+1)$ is $y = \cos^{-1} x$ shifted left 1 unit. It needs $-1 \leq x+1 \leq 1$, so the domain is $-2 \leq x \leq 0$. The range is $0 \leq y \leq \pi$. Key points: $(-2, \pi)$, $\left(-1, \frac{\pi}{2}\right)$, $(0, 0)$. The function is neither even nor odd.

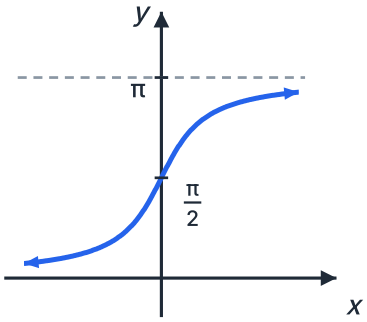
c $y = \frac{\pi}{2} + \tan^{-1} x$ is $y = \tan^{-1} x$ shifted up $\frac{\pi}{2}$ units. The domain is all real x, and the range is $0 < y < \pi$. The y-intercept is $\left(0, \frac{\pi}{2}\right)$. As $x \to \infty$, $y \to \pi$, and as $x \to -\infty$, $y \to 0$, so the asymptotes are $y = 0$ and $y = \pi$. The function is neither even nor odd: replacing x by $-x$ gives $\frac{\pi}{2} - \tan^{-1} x$, which is neither y nor $-y$.



a $y = \sin^{-1}(x-1)$



b $y = \cos^{-1}(x+1)$



c $y = \frac{\pi}{2} + \tan^{-1} x$

4 Exercise 17E Q8

a Let $\alpha = \cos^{-1}\frac{1}{3}$. Then $\cos\alpha = \frac{1}{3}$, where $0 \leq \alpha \leq \pi$.

$$\cos\left(2\cos^{-1}\tfrac{1}{3}\right) = \cos 2\alpha = 2\cos^2\alpha - 1 = 2 \times \tfrac{1}{9} - 1 = -\tfrac{7}{9}$$

b Let $\alpha = \cos^{-1}\frac{6}{7}$. Then $\cos\alpha = \frac{6}{7}$, where $0 \leq \alpha \leq \pi$, so α is in the first quadrant.

The third side of the triangle is $\sqrt{7^2 - 6^2} = \sqrt{13}$, so $\sin\alpha = \frac{\sqrt{13}}{7}$.

$$\sin\left(2\cos^{-1}\tfrac{6}{7}\right) = \sin 2\alpha = 2\sin\alpha\cos\alpha = 2 \times \tfrac{\sqrt{13}}{7} \times \tfrac{6}{7} = \tfrac{12\sqrt{13}}{49}$$

c Let $\alpha = \tan^{-1}(-2)$. Then $\tan\alpha = -2$, where $-\frac{\pi}{2} < \alpha < \frac{\pi}{2}$.

$$\tan\left(2\tan^{-1}(-2)\right) = \tan 2\alpha = \frac{2\tan\alpha}{1 - \tan^2\alpha} = \frac{2 \times (-2)}{1 - (-2)^2} = \frac{-4}{-3} = \frac{4}{3}$$

5 Standard integrals (Parts 1 and 2)

a $\displaystyle\int (4x^3 - 6x + 5)\,dx = 4 \times \frac{x^4}{4} - 6 \times \frac{x^2}{2} + 5x + C = x^4 - 3x^2 + 5x + C$

b $\displaystyle\int (2x+1)^4\,dx = \frac{(2x+1)^5}{2 \times 5} + C = \frac{(2x+1)^5}{10} + C \qquad (a = 2,\ n = 4)$

c $\displaystyle\int e^{-2x}\,dx = \frac{1}{-2}\,e^{-2x} + C = -\tfrac{1}{2}e^{-2x} + C \qquad (a = -2)$

d $\displaystyle\int \frac{1}{2x+5}\,dx = \tfrac{1}{2}\log_e|2x+5| + C \qquad (a = 2)$

e $\displaystyle\int \sin 4x\,dx = -\tfrac{1}{4}\cos 4x + C \qquad (a = 4)$

f $\displaystyle\int \cos \tfrac{1}{2}x\,dx = \frac{1}{\frac{1}{2}}\sin \tfrac{1}{2}x + C = 2\sin \tfrac{1}{2}x + C \qquad (a = \tfrac{1}{2})$

g $\displaystyle\int \sec^2 3x\,dx = \tfrac{1}{3}\tan 3x + C \qquad (a = 3)$

h Let $f(x) = x^2 + 1$. Then $f'(x) = 2x$, so the top is the derivative of the bottom:

$$\int \frac{2x}{x^2+1}\,dx = \log_e(x^2+1) + C$$

No absolute value is needed, because $x^2 + 1 > 0$ for all x.

i Let $f(x) = x^3 + 1$. Then $f'(x) = 3x^2$, so by the reverse chain rule with $n = 4$:

$$\int 3x^2(x^3+1)^4\,dx = \frac{(x^3+1)^5}{5} + C$$

6 Standard integrals (Parts 1 and 2)

a $\displaystyle\int_1^2 (3x^2 - 2x)\,dx = \Big[x^3 - x^2\Big]_1^2 = (8-4) - (1-1) = 4$

b $\displaystyle\int_0^1 e^{2x}\,dx = \tfrac{1}{2}\Big[e^{2x}\Big]_0^1 = \tfrac{1}{2}\big(e^2 - e^0\big) = \tfrac{1}{2}\big(e^2 - 1\big)$

c $\displaystyle\int_1^e \frac{3}{x}\,dx = 3\Big[\log_e|x|\Big]_1^e = 3\big(\log_e e - \log_e 1\big) = 3(1-0) = 3$

d $\displaystyle\int_0^{\frac{\pi}{6}} \cos 2x\,dx = \tfrac{1}{2}\Big[\sin 2x\Big]_0^{\frac{\pi}{6}} = \tfrac{1}{2}\big(\sin \tfrac{\pi}{3} - \sin 0\big) = \tfrac{1}{2}\Big(\tfrac{\sqrt{3}}{2} - 0\Big) = \tfrac{\sqrt{3}}{4}$

e $\displaystyle\int_0^{\frac{\pi}{8}} \sec^2 2x\,dx = \tfrac{1}{2}\Big[\tan 2x\Big]_0^{\frac{\pi}{8}} = \tfrac{1}{2}\big(\tan \tfrac{\pi}{4} - \tan 0\big) = \tfrac{1}{2}(1-0) = \tfrac{1}{2}$

f $\displaystyle\int_0^{\frac{\pi}{3}} \sin x\,dx = \Big[-\cos x\Big]_0^{\frac{\pi}{3}} = -\cos \tfrac{\pi}{3} + \cos 0 = -\tfrac{1}{2} + 1 = \tfrac{1}{2}$

Integration Part 3 Further Integration - Homework Answers

Lesson 12A | Inverse trigonometric functions: differentiating

Lesson Examples

C1 **Tutor demo.**

a Let $u = x^2$ and $v = \cos^{-1} x$. Then $u' = 2x$ and $v' = -\dfrac{1}{\sqrt{1-x^2}}$.

$$\begin{aligned} y' &= vu' + uv' \\ &= \cos^{-1} x \times 2x + x^2 \times \left(-\frac{1}{\sqrt{1-x^2}}\right) \\ &= 2x\cos^{-1} x - \frac{x^2}{\sqrt{1-x^2}} \end{aligned}$$

b Let $u = ax + b$, then $y = \cos^{-1} u$. Hence $\dfrac{du}{dx} = a$ and $\dfrac{dy}{du} = -\dfrac{1}{\sqrt{1-u^2}}$.

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = -\frac{1}{\sqrt{1-(ax+b)^2}} \times a = -\frac{a}{\sqrt{1-(ax+b)^2}}$$

Your turn.

a Let $u = x^2$ and $v = \sin^{-1} x$. Then $u' = 2x$ and $v' = \dfrac{1}{\sqrt{1-x^2}}$.

$$\begin{aligned} y' &= vu' + uv' \\ &= \sin^{-1} x \times 2x + x^2 \times \frac{1}{\sqrt{1-x^2}} \\ &= 2x\sin^{-1} x + \frac{x^2}{\sqrt{1-x^2}} \end{aligned}$$

b Let $u = ax + b$, then $y = \tan^{-1} u$. Hence $\dfrac{du}{dx} = a$ and $\dfrac{dy}{du} = \dfrac{1}{1+u^2}$.

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = \frac{1}{1+(ax+b)^2} \times a = \frac{a}{1+(ax+b)^2}$$

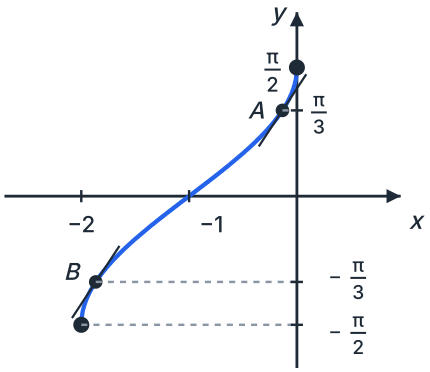
C2 **Tutor demo.**

a Differentiating, $y' = \dfrac{1}{\sqrt{1-(x+1)^2}}$.

Put $y' = 2$. Then

b The domain is $-1 \le x + 1 \le 1$, that is, $-2 \le x \le 0$, and the range is $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$. The endpoints are $\left(-2, -\frac{\pi}{2}\right)$ and $\left(0, \frac{\pi}{2}\right)$, and the curve crosses the x-axis at $x = -1$.

$$\frac{1}{\sqrt{1-(x+1)^2}} = 2$$
$$\sqrt{1-(x+1)^2} = \tfrac{1}{2}$$
$$1-(x+1)^2 = \tfrac{1}{4}$$
$$(x+1)^2 = \tfrac{3}{4}$$
$$x+1 = \tfrac{1}{2}\sqrt{3} \text{ or } -\tfrac{1}{2}\sqrt{3}$$
$$x = -1+\tfrac{1}{2}\sqrt{3} \text{ or } -1-\tfrac{1}{2}\sqrt{3}$$



When $x = -1+\tfrac{1}{2}\sqrt{3}$,
$y = \sin^{-1}\tfrac{1}{2}\sqrt{3} = \tfrac{\pi}{3}$.
When $x = -1-\tfrac{1}{2}\sqrt{3}$,
$y = \sin^{-1}\left(-\tfrac{1}{2}\sqrt{3}\right) = -\tfrac{\pi}{3}$.
So the points are $A\left(-1+\tfrac{1}{2}\sqrt{3}, \tfrac{\pi}{3}\right)$ and $B\left(-1-\tfrac{1}{2}\sqrt{3}, -\tfrac{\pi}{3}\right)$.

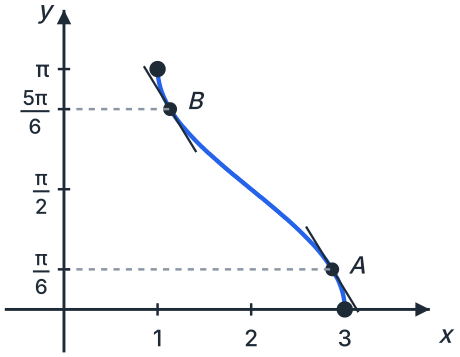
Your turn.

a Differentiating, $y' = -\dfrac{1}{\sqrt{1-(x-2)^2}}$.
Put $y' = -2$. Then

$$-\frac{1}{\sqrt{1-(x-2)^2}} = -2$$
$$\sqrt{1-(x-2)^2} = \tfrac{1}{2}$$
$$1-(x-2)^2 = \tfrac{1}{4}$$
$$(x-2)^2 = \tfrac{3}{4}$$
$$x-2 = \tfrac{1}{2}\sqrt{3} \text{ or } -\tfrac{1}{2}\sqrt{3}$$
$$x = 2+\tfrac{1}{2}\sqrt{3} \text{ or } 2-\tfrac{1}{2}\sqrt{3}$$

When $x = 2+\tfrac{1}{2}\sqrt{3}$,
$y = \cos^{-1}\tfrac{1}{2}\sqrt{3} = \tfrac{\pi}{6}$.
When $x = 2-\tfrac{1}{2}\sqrt{3}$,
$y = \cos^{-1}\left(-\tfrac{1}{2}\sqrt{3}\right) = \tfrac{5\pi}{6}$.
So the points are $A\left(2+\tfrac{1}{2}\sqrt{3}, \tfrac{\pi}{6}\right)$ and $B\left(2-\tfrac{1}{2}\sqrt{3}, \tfrac{5\pi}{6}\right)$.

b The domain is $-1 \le x-2 \le 1$, that is, $1 \le x \le 3$, and the range is $0 \le y \le \pi$. The endpoints are $(1, \pi)$ and $(3, 0)$, and the curve passes through $\left(2, \tfrac{\pi}{2}\right)$.



C3 **Tutor demo.**

a By Box 2 with $a = -1$ and $b = 0$, $\dfrac{d}{dx}\sin^{-1}(-x) = \dfrac{-1}{\sqrt{1-(-x)^2}} = \dfrac{-1}{\sqrt{1-x^2}}$.

So

$$\frac{d}{dx}\left(\sin^{-1}x + \sin^{-1}(-x)\right) = \frac{1}{\sqrt{1-x^2}} + \frac{-1}{\sqrt{1-x^2}} = 0$$

b Hence $\sin^{-1}x + \sin^{-1}(-x) = C$, for some constant C. Substitute $x = 0$, then $0 + 0 = C$, so $C = 0$. Thus $\sin^{-1}x + \sin^{-1}(-x) = 0$, that is, $\sin^{-1}(-x) = -\sin^{-1}x$, as required.

Your turn.

a By Box 2 with $a = -1$ and $b = 0$, $\dfrac{d}{dx}\tan^{-1}(-x) = \dfrac{-1}{1+(-x)^2} = \dfrac{-1}{1+x^2}$. So

$$\frac{d}{dx}\left(\tan^{-1}x + \tan^{-1}(-x)\right) = \frac{1}{1+x^2} + \frac{-1}{1+x^2} = 0$$

b Hence $\tan^{-1}x + \tan^{-1}(-x) = C$, for some constant C. Substitute $x = 0$, then $0 + 0 = C$, so $C = 0$. Thus $\tan^{-1}(-x) = -\tan^{-1}x$, as required.

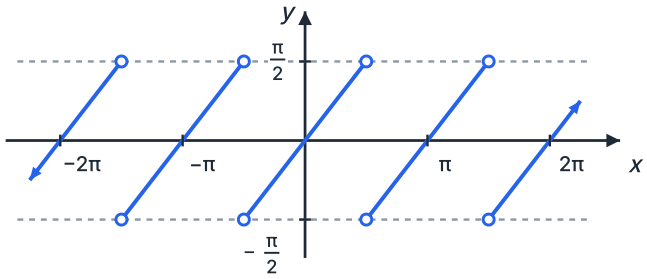
C4 **Tutor demo.** $\tan x$ is undefined when $x = \frac{\pi}{2} + k\pi$, for every integer k, so y is undefined there too. The function is periodic with the same period π as $\tan x$. A table of test values:

x	$-\frac{\pi}{4}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$
y	$-\frac{\pi}{4}$	0	$\frac{\pi}{4}$	undefined	$-\frac{\pi}{4}$	0	$\frac{\pi}{4}$

Let $u = \tan x$, then $y = \tan^{-1}u$. Hence $\dfrac{du}{dx} = \sec^2 x$ and $\dfrac{dy}{du} = \dfrac{1}{1+u^2}$.

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = \frac{1}{1+\tan^2 x} \times \sec^2 x = \frac{\sec^2 x}{\sec^2 x} = 1$$

So the graph is a series of separate intervals, each with gradient 1. Each runs from $\left(k\pi - \frac{\pi}{2}, -\frac{\pi}{2}\right)$ to $\left(k\pi + \frac{\pi}{2}, \frac{\pi}{2}\right)$ through $(k\pi, 0)$, and the endpoints are not included.



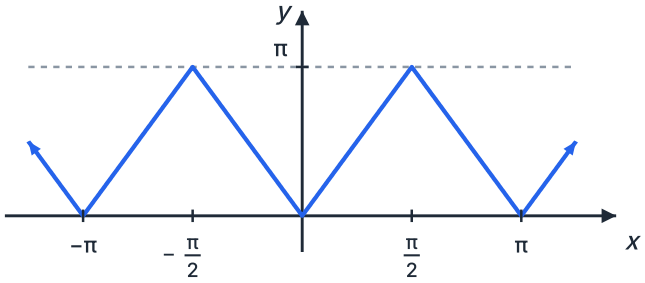
Your turn. The function is periodic with the same period π as $\cos 2x$. A table of test values:

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$
y	0	$\frac{\pi}{2}$	π	$\frac{\pi}{2}$	0	$\frac{\pi}{2}$	π

Let $u = \cos 2x$, then $y = \cos^{-1} u$. Hence $\dfrac{du}{dx} = -2\sin 2x$ and $\dfrac{dy}{du} = -\dfrac{1}{\sqrt{1-u^2}}$.

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \times \frac{du}{dx} = -\frac{1}{\sqrt{1-\cos^2 2x}} \times (-2\sin 2x) \\ &= \frac{2\sin 2x}{\sqrt{1-\cos^2 2x}} = \frac{2\sin 2x}{\sqrt{\sin^2 2x}}\end{aligned}$$

When $\sin 2x$ is positive, $\sqrt{\sin^2 2x} = \sin 2x$, so $\dfrac{dy}{dx} = 2$. When $\sin 2x$ is negative, $\sqrt{\sin^2 2x} = -\sin 2x$, so $\dfrac{dy}{dx} = -2$. So $\dfrac{dy}{dx} = 2$ for $0 < x < \frac{\pi}{2}$ and $\dfrac{dy}{dx} = -2$ for $\frac{\pi}{2} < x < \pi$, and this repeats every π. The graph is a zigzag of intervals with gradient 2 or -2, between $y = 0$ and $y = \pi$.



C5 **Tutor demo.**

a Here $u = e^{-2x}$ and $u' = -2e^{-2x}$, so

$$\frac{d}{dx}\left(\tan^{-1} e^{-2x}\right) = \frac{u'}{1+u^2} = \frac{-2e^{-2x}}{1+e^{-4x}}.$$

b Here $u = x^2 - 1$ and $u' = 2x$. Also $1 - u^2 = 1 - (x^4 - 2x^2 + 1) = 2x^2 - x^4$. So

$$\frac{d}{dx}\cos^{-1}(x^2-1) = -\frac{u'}{\sqrt{1-u^2}} = -\frac{2x}{\sqrt{1-(x^2-1)^2}} = -\frac{2x}{\sqrt{2x^2-x^4}}.$$

Your turn.

a Here $u = e^{-4x}$ and $u' = -4e^{-4x}$, so

$$\frac{d}{dx}\left(\tan^{-1} e^{-4x}\right) = \frac{u'}{1+u^2} = \frac{-4e^{-4x}}{1+e^{-8x}}.$$

b Here $u = 1 - x^2$ and $u' = -2x$. Also $1 - u^2 = 1 - (1 - 2x^2 + x^4) = 2x^2 - x^4$. So

$$\frac{d}{dx}\sin^{-1}(1-x^2) = \frac{u'}{\sqrt{1-u^2}} = \frac{-2x}{\sqrt{1-(1-x^2)^2}} = \frac{-2x}{\sqrt{2x^2-x^4}}.$$

Textbook Homework

1 From $\dfrac{dy}{dx} = \dfrac{1}{\sqrt{1-x^2}}$ (your measured values should be close to these):

x	-1	-0.7	-0.5	-0.2	0	0.3	0.6	0.8	1
$\dfrac{dy}{dx}$	vertical	1.40	1.15	1.02	1.00	1.05	1.25	1.67	vertical

At $x = \pm 1$ the tangent is vertical, so the gradient is undefined.

2

a $\dfrac{-1}{\sqrt{1-x^2}}$ **b** $\dfrac{1}{1+x^2}$ **c** $\dfrac{2}{\sqrt{1-4x^2}}$

d $\dfrac{3}{1+9x^2}$ **e** $\dfrac{-5}{\sqrt{1-25x^2}}$ **f** $\dfrac{-1}{\sqrt{1-x^2}}$

g $\dfrac{2x}{\sqrt{1-x^4}}$ **h** $\dfrac{3x^2}{1+x^6}$ **i** $\dfrac{1}{x^2+4x+5}$

j $\dfrac{1}{\sqrt{2x-x^2}}$ **k** $\sin^{-1} x + \dfrac{x}{\sqrt{1-x^2}}$ **l** $2x\tan^{-1} x + 1$

m $\dfrac{1}{\sqrt{25-x^2}}$ **n** $\dfrac{4}{16+x^2}$ **o** $\dfrac{-1}{2\sqrt{x-x^2}}$

p $\dfrac{1}{2(1+x)\sqrt{x}}$ **q** $\dfrac{-1}{1+x^2}$

3 **a** 2 **b** 2 **c** 1 **d** -1

4 **a** Tangent is $y = -6x + \pi$, normal is $y = \frac{1}{6}x + \pi$.

b Tangent is $y = \frac{1}{\sqrt{2}}x + \frac{\pi}{4} - 1$, normal is $y = -\sqrt{2}\,x + \frac{\pi}{4} + 2$.

5 **b** $\frac{\pi}{2}$

6 **a** π **b** $\frac{\pi}{2}$

7 **b** concave up

9 **a** $\cos^{-1} x$ **b** $\dfrac{3e^{3x}}{\sqrt{1-e^{6x}}}$ **c** $\dfrac{2}{\sqrt{7+12x-4x^2}}$

d $\dfrac{1}{x^2-2x+2}$ **e** $\dfrac{e^x}{\sqrt{1-e^{2x}}}$ **f** $\dfrac{1}{2\sqrt{1-x^2}\,\sin^{-1} x}$

g $\dfrac{1}{2x\sqrt{\log_e x\,(1-\log_e x)}}$ **h** $\dfrac{\sin^{-1}\sqrt{1-x}}{2\sqrt{x}} - \dfrac{1}{2\sqrt{1-x}}$

i $\dfrac{1}{1+x^2}$

11 **a** $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$ **b** $x = \frac{1}{2}\sin y$

c $\dfrac{dx}{dy} = \frac{1}{2}\cos y \geq 0$ for $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$ **d** $\dfrac{dy}{dx} = \dfrac{2}{\sqrt{1-4x^2}}$

12 **a** $\dfrac{1}{\sqrt{4-x^2}}$ **b** $\dfrac{-1}{\sqrt{2x-x^2}}$ **c** $\dfrac{1}{2(1+x)\sqrt{x}}$

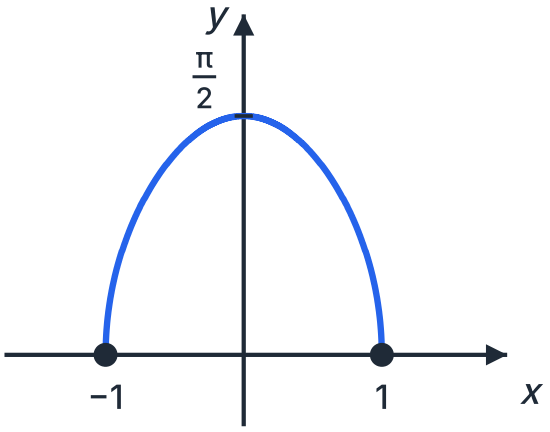
13 **a** $-1 \leq x \leq 1$, even

b The y-axis, because the function is even.

c $\dfrac{-2x}{\sqrt{1-x^4}}$

e The tangents at $x = 1$ and $x = -1$ are vertical.

f



15 **c** $\frac{1}{45}$ rad/s

16 **a** $x \neq 0$, odd

17 **a** $x \geq 1$ or $x \leq -1$

c They are undefined.

d When $x > 1$, $f'(x) = \dfrac{1}{x\sqrt{x^2-1}}$,

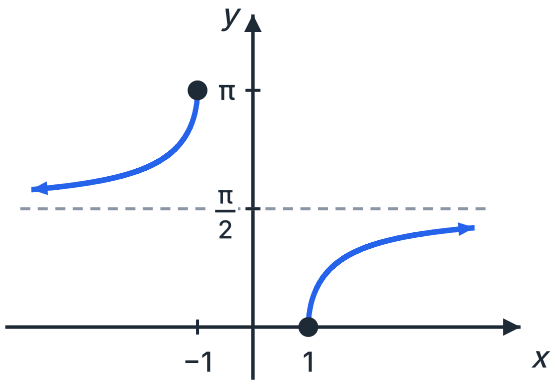
and when $x < -1$, $f'(x) = \dfrac{-1}{x\sqrt{x^2-1}}$

.

e $f'(x) > 0$ for $x > 1$ and for $x < -1$.

f (i) $\frac{\pi}{2}$ (ii) $\frac{\pi}{2}$

g



18 **a** $\dfrac{3e^{3x}}{1+e^{6x}}$ **b** $\dfrac{3x^2}{\sqrt{1-x^6}}$ **c** $-\dfrac{1}{x\sqrt{1-(\log_e x)^2}}$

19 **a** $-1 \leq x \leq 1$ **c** $g(x) = \frac{\pi}{2}$ for $0 \leq x \leq 1$.

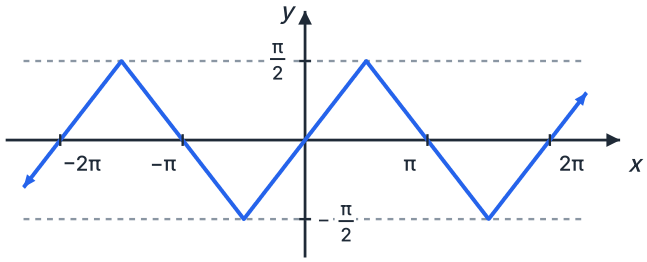
20 $\tan^{-1}\dfrac{x+2}{1-2x}$ is $\tan^{-1}x + \tan^{-1}2$ for $x < \frac{1}{2}$, and is $\tan^{-1}x + \tan^{-1}2 - \pi$ for $x > \frac{1}{2}$.

21 **a** domain: all real x, range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$, odd

c No, because $\frac{0}{0}$ is undefined.

d $f'(x) = 1$ when $\cos x > 0$, and $f'(x) = -1$ when $\cos x < 0$.

e The graph is a zigzag of intervals with gradient 1 or -1, between $y = -\frac{\pi}{2}$ and $y = \frac{\pi}{2}$, through $(k\pi, 0)$.



Integration Part 3 Further Integration - Homework Answers

Lesson 12A | Inverse trigonometric functions: differentiating

Past-Paper Homework - Worked Solutions

PP 1 **Cherrybrook Technology High School, 2020, Year 12 AP4, Mathematics Extension 1, Q4**

By Box 2 with $a = 3$ and $b = 0$,

$$\frac{d}{dx}\sin^{-1} 3x = \frac{3}{\sqrt{1-(3x)^2}} = \frac{3}{\sqrt{1-9x^2}}.$$

Answer B. Options A, C and D have the minus sign of the $\cos^{-1}$ form, A and C also leave out the factor 3 from the chain rule, and C squares only x, not $3x$.

PP 2 **Baulkham Hills High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q9**

For $x\cos^{-1} x$ use the product rule with $u = x$, $v = \cos^{-1} x$, $u' = 1$, $v' = -\dfrac{1}{\sqrt{1-x^2}}$.

For $\sqrt{1-x^2} = (1-x^2)^{\frac{1}{2}}$ use the chain rule: its derivative is $\frac{1}{2}(1-x^2)^{-\frac{1}{2}} \times (-2x) = \dfrac{-x}{\sqrt{1-x^2}}$. So

$$\begin{aligned}\frac{d}{dx}\left(x\cos^{-1} x - \sqrt{1-x^2}\right) &= \cos^{-1} x \times 1 + x \times \left(-\frac{1}{\sqrt{1-x^2}}\right) - \frac{-x}{\sqrt{1-x^2}} \\ &= \cos^{-1} x - \frac{x}{\sqrt{1-x^2}} + \frac{x}{\sqrt{1-x^2}} \\ &= \cos^{-1} x.\end{aligned}$$

Answer C.

PP 3 **Sydney Grammar School, 2022, Trial Examination, Form VI Mathematics Extension 1, Q13(a)**

Here $u = 3x^2$ and $u' = 6x$. By Box 3,

$$\frac{d}{dx}\left(\cos^{-1}(3x^2)\right) = -\frac{u'}{\sqrt{1-u^2}} = -\frac{1}{\sqrt{1-(3x^2)^2}} \times 6x = -\frac{6x}{\sqrt{1-9x^4}}.$$

PP 4 **Sydney Technical High School, 2021, Extension 1 Mathematics, Trial HSC, Q11(g)**

Let $u = e^{-x}$ and $v = \sin^{-1} x$. Then $u' = -e^{-x}$ and $v' = \dfrac{1}{\sqrt{1-x^2}}$. By the product rule,

$$\begin{aligned}\frac{dy}{dx} &= vu' + uv' \\ &= \sin^{-1} x \times \left(-e^{-x}\right) + e^{-x} \times \frac{1}{\sqrt{1-x^2}} \\ &= -e^{-x}\sin^{-1} x + \frac{e^{-x}}{\sqrt{1-x^2}}.\end{aligned}$$

PP 5 **Abbotsleigh, 2024, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q13(d)**

When $x = -\frac{\pi}{2}$, $y = \sin^{-1}\left(-\frac{\pi}{2} + \frac{\pi}{2}\right) + \frac{\pi}{2} = \sin^{-1} 0 + \frac{\pi}{2} = \frac{\pi}{2}$. So the point is $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

By Box 2 with $a = 1$ and $b = \frac{\pi}{2}$, $\dfrac{dy}{dx} = \dfrac{1}{\sqrt{1-\left(x+\frac{\pi}{2}\right)^2}}$. At $x = -\frac{\pi}{2}$,

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-0^2}} = 1.$$

So the normal has gradient -1, and its equation is

$$\begin{aligned}y - \tfrac{\pi}{2} &= -1\left(x + \tfrac{\pi}{2}\right) \\ y &= -x - \tfrac{\pi}{2} + \tfrac{\pi}{2} \\ y &= -x.\end{aligned}$$

When $x = 0$, $y = 0$, so the normal passes through the origin.

PP 6 **Hurlstone Agricultural High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q14(a)(i)(ii)**

(i) Let $u = x$ and $v = \tan^{-1} x$. Then $u' = 1$ and $v' = \dfrac{1}{1+x^2}$. By the product rule,

$$\frac{d}{dx}\left(x\tan^{-1}x\right) = \tan^{-1}x \times 1 + x \times \frac{1}{1+x^2} = \tan^{-1}x + \frac{x}{1+x^2}.$$

(ii) From (i), $\tan^{-1}x = \dfrac{d}{dx}\left(x\tan^{-1}x\right) - \dfrac{x}{1+x^2}$. Integrate both sides from 0 to $\sqrt{3}$. For the last term, $\displaystyle\int \frac{x}{1+x^2}\,dx = \tfrac{1}{2}\int \frac{2x}{1+x^2}\,dx = \tfrac{1}{2}\ln(1+x^2) + C$, since $2x$ is the derivative of $1+x^2$.

$$\begin{aligned}
\int_0^{\sqrt{3}} \tan^{-1}x\,dx &= \left[x\tan^{-1}x\right]_0^{\sqrt{3}} - \int_0^{\sqrt{3}} \frac{x}{1+x^2}\,dx \\
&= \left(\sqrt{3}\tan^{-1}\sqrt{3} - 0\right) - \left[\tfrac{1}{2}\ln(1+x^2)\right]_0^{\sqrt{3}} \\
&= \sqrt{3} \times \frac{\pi}{3} - \left(\tfrac{1}{2}\ln 4 - \tfrac{1}{2}\ln 1\right) \\
&= \frac{\pi\sqrt{3}}{3} - \tfrac{1}{2}\ln 4 \\
&= \frac{\pi\sqrt{3}}{3} - \ln 2.
\end{aligned}$$

PP 7 **Blacktown Boys' High School, 2024, HSC Trial Examination, Mathematics Extension 1, Q11(d)**

Use the quotient rule with $u = \sin^{-1}(3x)$ and $v = 3x$. Then $u' = \dfrac{3}{\sqrt{1-9x^2}}$ (Box 2 with $a = 3$) and $v' = 3$. In the third line, the top and bottom are multiplied by $\sqrt{1-9x^2}$.

$$\begin{aligned}
\frac{d}{dx}\left(\frac{\sin^{-1}(3x)}{3x}\right) &= \frac{vu' - uv'}{v^2} \\
&= \frac{3x \times \dfrac{3}{\sqrt{1-9x^2}} - \sin^{-1}(3x) \times 3}{9x^2} \\
&= \frac{9x - 3\sin^{-1}(3x)\sqrt{1-9x^2}}{9x^2\sqrt{1-9x^2}} \\
&= \frac{3\left(3x - \sin^{-1}(3x)\sqrt{1-9x^2}\right)}{9x^2\sqrt{1-9x^2}} \\
&= \frac{3x - \sin^{-1}(3x)\sqrt{1-9x^2}}{3x^2\sqrt{1-9x^2}}.
\end{aligned}$$

PP 8 **Knox Grammar School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q12(c)**

By the chain rule, with Box 2 for $\sin^{-1}\dfrac{x}{b}$ (here $a=\frac{1}{b}$),

$$g'(x)=2\sin^{-1}\frac{x}{b}\times\frac{d}{dx}\left(\sin^{-1}\frac{x}{b}\right)=2\sin^{-1}\frac{x}{b}\times\frac{\frac{1}{b}}{\sqrt{1-\frac{x^2}{b^2}}}.$$

At $x=\frac{b}{2}$, $\dfrac{x}{b}=\frac{1}{2}$, so $\sin^{-1}\dfrac{x}{b}=\sin^{-1}\frac{1}{2}=\frac{\pi}{6}$ and $\sqrt{1-\dfrac{x^2}{b^2}}=\sqrt{1-\frac{1}{4}}=\dfrac{\sqrt{3}}{2}$.

Hence

$$g'\left(\frac{b}{2}\right)=2\times\frac{\pi}{6}\times\frac{1}{b}\times\frac{2}{\sqrt{3}}=\frac{2\pi}{3\sqrt{3}\,b}.$$

This gradient is $\dfrac{2\pi}{3}$, so

$$\frac{2\pi}{3\sqrt{3}\,b}=\frac{2\pi}{3},\quad\text{so}\quad\sqrt{3}\,b=1,\quad\text{and}\quad b=\frac{1}{\sqrt{3}}=\frac{\sqrt{3}}{3}.$$

PP 9 **Ascham School, 2020, Trial HSC Examination, Mathematics Extension 1, Q14(c)**

Work with $0<x<1$, where both derivatives exist. By Box 2 with $a=2$, $\dfrac{d}{dx}\cos^{-1}(2x-1)=-\dfrac{2}{\sqrt{1-(2x-1)^2}}$. By Box 3 with $u=\sqrt{x}$ and $u'=\dfrac{1}{2\sqrt{x}}$,

$$\frac{d}{dx}\cos^{-1}\sqrt{x}=-\frac{u'}{\sqrt{1-u^2}}=-\frac{\frac{1}{2\sqrt{x}}}{\sqrt{1-x}}=-\frac{1}{2\sqrt{x}\,\sqrt{1-x}}.$$

Also $1-(2x-1)^2=1-(4x^2-4x+1)=4x-4x^2=4x(1-x)$, so $\sqrt{1-(2x-1)^2}=2\sqrt{x}\,\sqrt{1-x}$. Hence

$$\begin{aligned}f'(x)&=-\frac{2}{\sqrt{1-(2x-1)^2}}-2\times\left(-\frac{1}{2\sqrt{x}\,\sqrt{1-x}}\right)\\&=-\frac{2}{2\sqrt{x}\,\sqrt{1-x}}+\frac{1}{\sqrt{x}\,\sqrt{1-x}}\\&=-\frac{1}{\sqrt{x}\,\sqrt{1-x}}+\frac{1}{\sqrt{x}\,\sqrt{1-x}}\\&=0.\end{aligned}$$

Integration Part 3 Further Integration - Homework Answers

Lesson 12B | Inverse trigonometric functions: integrating

Lesson Examples

C1 **Tutor demo.**

$$\begin{aligned}\int_0^{\frac{\sqrt{3}}{2}} \frac{1}{\sqrt{1-x^2}}\,dx &= \left[\sin^{-1}x\right]_0^{\frac{\sqrt{3}}{2}} \\ &= \sin^{-1}\frac{\sqrt{3}}{2} - \sin^{-1}0 \\ &= \frac{\pi}{3} - 0 \\ &= \frac{\pi}{3}\end{aligned}$$

$$\begin{aligned}\text{OR} \quad \int_0^{\frac{\sqrt{3}}{2}} \frac{1}{\sqrt{1-x^2}}\,dx &= \left[-\cos^{-1}x\right]_0^{\frac{\sqrt{3}}{2}} \\ &= -\cos^{-1}\frac{\sqrt{3}}{2} + \cos^{-1}0 \\ &= -\frac{\pi}{6} + \frac{\pi}{2} \\ &= \frac{\pi}{3}\end{aligned}$$

Your turn.

$$\begin{aligned}\int_0^{\frac{1}{\sqrt{2}}} \frac{1}{\sqrt{1-x^2}}\,dx &= \left[\sin^{-1}x\right]_0^{\frac{1}{\sqrt{2}}} \\ &= \sin^{-1}\frac{1}{\sqrt{2}} - \sin^{-1}0 \\ &= \frac{\pi}{4} - 0 \\ &= \frac{\pi}{4}\end{aligned}$$

$$\begin{aligned}\text{OR} \quad \int_0^{\frac{1}{\sqrt{2}}} \frac{1}{\sqrt{1-x^2}}\,dx &= \left[-\cos^{-1}x\right]_0^{\frac{1}{\sqrt{2}}} \\ &= -\cos^{-1}\frac{1}{\sqrt{2}} + \cos^{-1}0 \\ &= -\frac{\pi}{4} + \frac{\pi}{2} \\ &= \frac{\pi}{4}\end{aligned}$$

C2 Tutor demo.

a

$$\begin{aligned}\int_0^{\sqrt{3}} \frac{1}{1+x^2}\,dx &= \Big[\tan^{-1} x\Big]_0^{\sqrt{3}} \\ &= \tan^{-1}\sqrt{3} - \tan^{-1} 0 \\ &= \tfrac{\pi}{3} - 0 \\ &= \tfrac{\pi}{3}\end{aligned}$$

b

$$\begin{aligned}\int_0^{3} \frac{1}{1+x^2}\,dx &= \Big[\tan^{-1} x\Big]_0^{3} \\ &= \tan^{-1} 3 - \tan^{-1} 0 \\ &= 1.2490\ldots - 0 \\ &\doteq 1.249\end{aligned}$$

Your turn.

a

$$\begin{aligned}\int_{-1}^{1} \frac{1}{1+x^2}\,dx &= \Big[\tan^{-1} x\Big]_{-1}^{1} \\ &= \tan^{-1} 1 - \tan^{-1}(-1) \\ &= \tfrac{\pi}{4} - \left(-\tfrac{\pi}{4}\right) \\ &= \tfrac{\pi}{2}\end{aligned}$$

b

$$\begin{aligned}\int_0^{2} \frac{1}{1+x^2}\,dx &= \Big[\tan^{-1} x\Big]_0^{2} \\ &= \tan^{-1} 2 - \tan^{-1} 0 \\ &= 1.1071\ldots - 0 \\ &\doteq 1.107\end{aligned}$$

C3 **Tutor demo.**

a Here $a = 4$.

$$\begin{aligned}\int \frac{3}{\sqrt{16-x^2}}\,dx &= 3\int \frac{1}{\sqrt{4^2-x^2}}\,dx \\ &= 3\sin^{-1}\frac{x}{4} + C\end{aligned}$$

b Here $a = \sqrt{12} = 2\sqrt{3}$.

$$\begin{aligned}\int \frac{1}{12+x^2}\,dx &= \int \frac{1}{(2\sqrt{3})^2+x^2}\,dx \\ &= \frac{1}{2\sqrt{3}}\tan^{-1}\frac{x}{2\sqrt{3}} + C\end{aligned}$$

Your turn.

a Here $a = 5$.

$$\begin{aligned}\int \frac{5}{\sqrt{25-x^2}}\,dx &= 5\int \frac{1}{\sqrt{5^2-x^2}}\,dx \\ &= 5\sin^{-1}\frac{x}{5} + C\end{aligned}$$

b Here $a = \sqrt{18} = 3\sqrt{2}$.

$$\begin{aligned}\int \frac{1}{18+x^2}\,dx &= \int \frac{1}{(3\sqrt{2})^2+x^2}\,dx \\ &= \frac{1}{3\sqrt{2}}\tan^{-1}\frac{x}{3\sqrt{2}} + C\end{aligned}$$

C4 **Tutor demo.**

c Take out 16. Then $a^2 = \frac{9}{16}$, so $a = \frac{3}{4}$.

$$\int \frac{10}{9+16x^2}\,dx = \frac{10}{16}\int \frac{1}{\frac{9}{16}+x^2}\,dx$$
$$= \frac{5}{8} \times \frac{4}{3}\tan^{-1}\frac{x}{3/4} + C$$
$$= \frac{5}{6}\tan^{-1}\frac{4x}{3} + C$$

Box 6 with $a = 3, b = 4$: $10 \times \frac{1}{12}\tan^{-1}\frac{4x}{3} + C$, the same.

d Take out $\sqrt{2}$. Then $a^2 = \frac{7}{2}$, so $a = \sqrt{\frac{7}{2}}$.

$$\int \frac{1}{\sqrt{7-2x^2}}\,dx = \frac{1}{\sqrt{2}}\int \frac{1}{\sqrt{\frac{7}{2}-x^2}}\,dx$$
$$= \frac{1}{\sqrt{2}}\sin^{-1}\frac{x}{\sqrt{7/2}} + C$$
$$= \tfrac{1}{2}\sqrt{2}\sin^{-1}x\sqrt{\tfrac{2}{7}} + C$$

Box 6 with $a = \sqrt{7}, b = \sqrt{2}$: $\frac{1}{\sqrt{2}}\sin^{-1}\frac{\sqrt{2}\,x}{\sqrt{7}} + C$, the same.

Your turn.

c Take out 4. Then $a^2 = \frac{25}{4}$, so $a = \frac{5}{2}$.

$$\int \frac{12}{25+4x^2}\,dx = \frac{12}{4}\int \frac{1}{\frac{25}{4}+x^2}\,dx$$
$$= 3 \times \frac{2}{5}\tan^{-1}\frac{x}{5/2} + C$$
$$= \frac{6}{5}\tan^{-1}\frac{2x}{5} + C$$

Box 6 with $a = 5, b = 2$: $12 \times \frac{1}{10}\tan^{-1}\frac{2x}{5} + C$, the same.

d Take out $\sqrt{5}$. Then $a^2 = \frac{3}{5}$, so $a = \sqrt{\frac{3}{5}}$.

$$\int \frac{1}{\sqrt{3-5x^2}}\,dx = \frac{1}{\sqrt{5}}\int \frac{1}{\sqrt{\frac{3}{5}-x^2}}\,dx$$
$$= \frac{1}{\sqrt{5}}\sin^{-1}\frac{x}{\sqrt{3/5}} + C$$
$$= \tfrac{1}{5}\sqrt{5}\sin^{-1}x\sqrt{\tfrac{5}{3}} + C$$

Box 6 with $a=\sqrt{3}, b=\sqrt{5}$: $\dfrac{1}{\sqrt{5}}\sin^{-1}\dfrac{\sqrt{5}\,x}{\sqrt{3}}+C$, the same.

C5 **Tutor demo.**

a Let $u=e^{2x}$, then $u'=2e^{2x}$. Note that $e^{4x}=\left(e^{2x}\right)^2$ and $e^{2x}=\frac{1}{2}\times 2e^{2x}$.

$$\int \frac{e^{2x}}{1+e^{4x}}\,dx=\tfrac{1}{2}\int \frac{2e^{2x}}{1+\left(e^{2x}\right)^2}\,dx=\tfrac{1}{2}\tan^{-1}e^{2x}+C$$

b Let $u=x^3$, then $u'=3x^2$. Note that $x^6=\left(x^3\right)^2$ and $x^2=\frac{1}{3}\times 3x^2$.

$$\int \frac{x^2}{\sqrt{1-x^6}}\,dx=\tfrac{1}{3}\int \frac{3x^2}{\sqrt{1-\left(x^3\right)^2}}\,dx=\tfrac{1}{3}\sin^{-1}x^3+C$$

Your turn.

a Let $u=\sin x$, then $u'=\cos x$. Note that $\sin^2 x=(\sin x)^2$.

$$\int \frac{\cos x}{1+\sin^2 x}\,dx=\int \frac{\cos x}{1+(\sin x)^2}\,dx=\tan^{-1}(\sin x)+C$$

b Let $u=x^2$, then $u'=2x$. Note that $x^4=\left(x^2\right)^2$ and $x=\frac{1}{2}\times 2x$.

$$\int \frac{x}{1+x^4}\,dx=\tfrac{1}{2}\int \frac{2x}{1+\left(x^2\right)^2}\,dx=\tfrac{1}{2}\tan^{-1}x^2+C$$

C6 **Tutor demo.**

a Let $y=x\cos^{-1}x$. By the product rule with $u=x$ and $v=\cos^{-1}x$, where $u'=1$ and $v'=\dfrac{-1}{\sqrt{1-x^2}}$,

$$\frac{dy}{dx}=1\times\cos^{-1}x+x\times\frac{-1}{\sqrt{1-x^2}}=\cos^{-1}x-\frac{x}{\sqrt{1-x^2}}.$$

Integrating both sides and rearranging,

$$\int \cos^{-1}x\,dx-\int\frac{x}{\sqrt{1-x^2}}\,dx=x\cos^{-1}x$$

$$\int \cos^{-1}x\,dx=x\cos^{-1}x+\int\frac{x}{\sqrt{1-x^2}}\,dx.$$

By the reverse chain rule with $u=1-x^2$ and $\dfrac{du}{dx}=-2x$,

$$\begin{aligned}\int\frac{x}{\sqrt{1-x^2}}\,dx&=-\tfrac{1}{2}\int\left(1-x^2\right)^{-\frac{1}{2}}(-2x)\,dx\\&=-\tfrac{1}{2}\times\left(1-x^2\right)^{\frac{1}{2}}\times 2\\&=-\sqrt{1-x^2},\end{aligned}$$

so $\int \cos^{-1} x\,dx = x\cos^{-1} x - \sqrt{1-x^2} + C$.

b The curve is above the x-axis for $0 \leq x < 1$, so the area is

$$\begin{aligned}\int_0^1 \cos^{-1} x\,dx &= \left[x\cos^{-1} x - \sqrt{1-x^2}\right]_0^1 \\ &= (1\times 0 - 0) - (0 - 1) \\ &= 1 \text{ square unit.}\end{aligned}$$

(At $x=1$: $\cos^{-1} 1 = 0$ and $\sqrt{1-1}=0$. At $x=0$: $0\times\cos^{-1}0 = 0$ and $\sqrt{1-0}=1$.)

Your turn.

a Let $y = x\sin^{-1}\dfrac{x}{2}$. By the product rule with $u = x$ and $v = \sin^{-1}\dfrac{x}{2}$, where $u' = 1$ and $v' = \dfrac{1}{\sqrt{4-x^2}}$ (Box 5 reversed),

$$\frac{dy}{dx} = 1\times\sin^{-1}\frac{x}{2} + x\times\frac{1}{\sqrt{4-x^2}} = \sin^{-1}\frac{x}{2} + \frac{x}{\sqrt{4-x^2}}.$$

Integrating both sides and rearranging,

$$\int \sin^{-1}\frac{x}{2}\,dx + \int \frac{x}{\sqrt{4-x^2}}\,dx = x\sin^{-1}\frac{x}{2}$$

$$\int \sin^{-1}\frac{x}{2}\,dx = x\sin^{-1}\frac{x}{2} - \int \frac{x}{\sqrt{4-x^2}}\,dx.$$

By the reverse chain rule with $u = 4-x^2$ and $\dfrac{du}{dx} = -2x$,

$$\begin{aligned}-\int \frac{x}{\sqrt{4-x^2}}\,dx &= \tfrac{1}{2}\int (4-x^2)^{-\frac{1}{2}}(-2x)\,dx \\ &= \tfrac{1}{2}\times(4-x^2)^{\frac{1}{2}}\times 2 \\ &= \sqrt{4-x^2},\end{aligned}$$

so $\int \sin^{-1}\dfrac{x}{2}\,dx = x\sin^{-1}\dfrac{x}{2} + \sqrt{4-x^2} + C$.

b The curve is above the x-axis for $0 < x \leq 2$, so the area is

$$\begin{aligned}\int_0^2 \sin^{-1}\frac{x}{2}\,dx &= \left[x\sin^{-1}\frac{x}{2} + \sqrt{4-x^2}\right]_0^2 \\ &= \left(2\times\tfrac{\pi}{2} + 0\right) - (0+2) \\ &= \pi - 2 \text{ square units.}\end{aligned}$$

(At $x=2$: $\sin^{-1}1 = \frac{\pi}{2}$ and $\sqrt{4-4}=0$. At $x=0$: $0\times\sin^{-1}0 = 0$ and $\sqrt{4-0}=2$.)

Textbook Homework

1 **a, b** i $\tan^{-1} 1 = \frac{\pi}{4} \doteq 0.79$ ii $\tan^{-1} 2 \doteq 1.11$
iii $\tan^{-1} \frac{1}{2} - \tan^{-1}\left(-\frac{1}{2}\right) \doteq 0.93$ iv $\tan^{-1}(-1) - \tan^{-1}(-3) \doteq 0.46$
Each little square has area $0.1 \times 0.1 = 0.01$, so the counts should be close to 79, 111, 93 and 46 squares.

2 **a** $\cos^{-1} x + C$ **b** $\sin^{-1} \frac{x}{2} + C$ **c** $\frac{1}{3}\tan^{-1} \frac{x}{3} + C$ **d** $\sin^{-1} \frac{3x}{2} + C$
e $\frac{1}{\sqrt{2}}\tan^{-1} \frac{x}{\sqrt{2}} + C$ **f** $\cos^{-1} \frac{x}{\sqrt{5}} + C$

3 **a** $\frac{\pi}{2}$ **b** $\frac{\pi}{8}$ **c** $\frac{\pi}{4}$ **d** $\frac{\pi}{12}$ **e** $\frac{\pi}{6}$ **f** $\frac{5\pi}{12}$

4 **a** $y = \sin^{-1} x + \pi$ **b** $y = \tan^{-1} \frac{x}{4} + \frac{\pi}{4}$

5 **a** $\frac{\pi}{3}$ **b** $\frac{\pi}{4}$

6 **a** $\frac{1}{2}\sin^{-1} 2x + C$ **b** $\frac{1}{4}\tan^{-1} 4x + C$ **c** $\frac{1}{\sqrt{2}}\cos^{-1} \sqrt{2}\,x + C$ **d** $\frac{1}{3}\sin^{-1} \frac{3x}{2} + C$
e $\frac{1}{15}\tan^{-1} \frac{3x}{5} + C$ **f** $\frac{1}{2}\cos^{-1} \frac{2x}{\sqrt{3}} + C$

7 **a** $\frac{\pi}{18}$ **b** $\frac{\pi}{12}$ **c** $\frac{2\pi}{9}\sqrt{3}$ **d** $\frac{5\pi}{24}$

8 **c** $\left(\frac{\pi}{12} + \frac{1}{2}\sqrt{3} - 1\right)$ unit^2

9 **b** $\left(1 - \frac{1}{2}\sqrt{3}\right)$ unit^2
c The rectangle with corners $(0, 0)$ and $\left(\frac{1}{2}, \frac{\pi}{6}\right)$ has area $\frac{\pi}{12}$, and $\frac{\pi}{12} - \left(1 - \frac{1}{2}\sqrt{3}\right) = \frac{\pi}{12} + \frac{1}{2}\sqrt{3} - 1$.

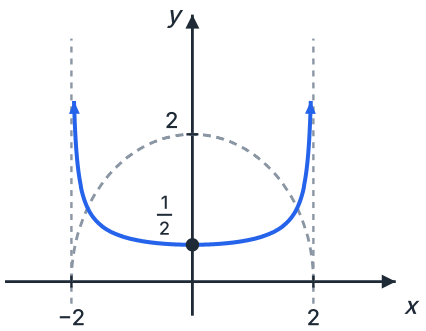
10 **b** $\frac{\pi}{2}$

11 **a** $\dfrac{6x^2}{4 + x^6}$ **b** $\frac{1}{6}\tan^{-1} \frac{x^3}{2} + C$

12 **a** $\tan^{-1} x + \dfrac{x}{1 + x^2}$ **b** $\frac{\pi}{4} - \frac{1}{2}\ln 2$

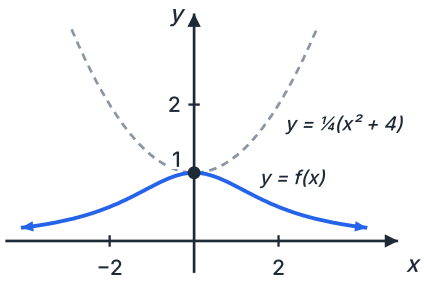
13 **a** 0 **b** 0 **c** $\frac{3\pi}{4}$ **d** 0 **e** 0 **f** 18π

14 **a** (i) 0
b (i) $f(0) = 0$ and $f'(x) < 0$ for $x > 0$, so $f(x)$ is decreasing for $x \geq 0$. (ii) $\frac{\pi - 2}{8}$

15 

a, b The semicircle $y = \sqrt{4 - x^2}$ is dashed. The graph of $y = f(x)$ is its reciprocal, with vertical asymptotes $x = 2$ and $x = -2$, and minimum $\left(0, \frac{1}{2}\right)$.
c domain: $-2 < x < 2$, range: $y \geq \frac{1}{2}$, even
d $\frac{\pi}{3}$ unit^2 **e** π unit^2

16



a The y-axis, because it is an even function.
b domain: all real x, range: $0 < y \leq 1$
d 0. The dashed curve is $y = \frac{1}{4}\left(x^2 + 4\right)$, the reciprocal of $f(x)$.
e π unit^2 **f** $4\tan^{-1}\frac{a}{2}$ unit^2 **g** 2π unit^2

18 **a** $\frac{5323}{6800}$

20 **a** $2\tan^{-1}\sqrt{x} + C$ **b** $\tan^{-1} e - \frac{\pi}{4}$

21 **b** The working gives $\frac{1}{6}\tan^{-1}\left(\frac{3}{2}\tan 7\right) \doteq 0.153$ unit^2.
c The integrand is well-defined in the interval $[0, 7]$, and lies between $\frac{1}{4}$ and $\frac{1}{9}$, so the area lies between $\frac{7}{4}$ and $\frac{7}{9}$, which is much larger than the answer of 0.153 that was calculated in part **b**. The primitive, however, is undefined at two values within the interval, at $x = \frac{\pi}{2}$ and at $x = \frac{3\pi}{2}$, which renders the argument completely invalid.

22 **a** $\dfrac{1 + t^{4n+2}}{1 + t^2}$ **b** $\dfrac{1 - t^{4n+4}}{1 + t^2}$ **g** $\pi \doteq 3.092$, error $\doteq 0.050$

23 **a** $S_n = \tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3 + \cdots + \tan^{-1} n$
b $x\tan^{-1} x - \frac{1}{2}\ln\left(1 + x^2\right)$

Integration Part 3 Further Integration - Homework Answers

Lesson 12B | Inverse trigonometric functions: integrating

Past-Paper Homework - Worked Solutions

PP 1 **Sydney Technical High School, 2021, Extension 1 Mathematics, Trial HSC, Q5**

Here $16 - x^2 = 4^2 - x^2$, so $a = 4$ in Box 5:

$$\int \frac{dx}{\sqrt{16 - x^2}} = \sin^{-1} \frac{x}{4} + c.$$

Options A and B have $4x$ in place of $\frac{x}{4}$. Option D has the wrong sign: the $\cos^{-1}$ form is $-\cos^{-1} \frac{x}{4} + c$. **Answer C.**

PP 2 **Knox Grammar School, 2022, Higher School Certificate Trial Examination, Mathematics Extension 1, Q1**

Here $2 + x^2 = (\sqrt{2})^2 + x^2$, so $a = \sqrt{2}$ in Box 5:

$$\begin{aligned}\int \frac{2}{2 + x^2}\, dx &= 2 \int \frac{1}{(\sqrt{2})^2 + x^2}\, dx \\ &= 2 \times \frac{1}{\sqrt{2}} \tan^{-1} \frac{x}{\sqrt{2}} + c \\ &= \sqrt{2} \tan^{-1} \frac{x}{\sqrt{2}} + c,\end{aligned}$$

because $\frac{2}{\sqrt{2}} = \sqrt{2}$. **Answer C.**

PP 3 **James Ruse Agricultural High School, 2020, Year 12 Trial HSC Examination, Mathematics Extension 1, Q11(a)**

Here $2 + x^2 = (\sqrt{2})^2 + x^2$, so $a = \sqrt{2}$ in Box 5:

$$\int \frac{5}{2 + x^2}\, dx = 5 \int \frac{1}{(\sqrt{2})^2 + x^2}\, dx = \frac{5}{\sqrt{2}} \tan^{-1} \frac{x}{\sqrt{2}} + C.$$

PP 4 **Moriah College, 2023, Trial Higher School Certificate Examination, Mathematics Extension 1, Q9(b)**

Here $a = 2$ in Box 5:

$$\begin{aligned}\int_0^{\sqrt{2}} \frac{dx}{\sqrt{4-x^2}} &= \left[\sin^{-1}\frac{x}{2}\right]_0^{\sqrt{2}} \\ &= \sin^{-1}\frac{\sqrt{2}}{2} - \sin^{-1} 0 \\ &= \sin^{-1}\frac{1}{\sqrt{2}} - 0 \\ &= \frac{\pi}{4}\end{aligned}$$

PP 5 **Blacktown Boys' High School, 2024, HSC Trial Examination, Mathematics Extension 1, Q12(a)**

Take out the coefficient 25 of x^2. Then $a^2 = \frac{1}{25}$, so $a = \frac{1}{5}$ and $\dfrac{1}{a} = 5$:

$$\begin{aligned}\int_0^{\frac{1}{5}} \frac{dx}{1+25x^2} &= \frac{1}{25}\int_0^{\frac{1}{5}} \frac{dx}{\frac{1}{25}+x^2} \\ &= \frac{1}{25} \times 5\left[\tan^{-1} 5x\right]_0^{\frac{1}{5}} \\ &= \frac{1}{5}\left(\tan^{-1} 1 - \tan^{-1} 0\right) \\ &= \frac{1}{5} \times \frac{\pi}{4} \\ &= \frac{\pi}{20}\end{aligned}$$

(By Box 6 with $a = 1$ and $b = 5$, the primitive is $\frac{1}{5}\tan^{-1} 5x$ at once.)

PP 6 **St George Girls High School, 2020, Trial HSC Examination, Mathematics Extension 1, Q15(a)**

Here $4x^2 = (2x)^2$. By Box 6 with $a = 1$ and $b = 2$, the primitive of $\dfrac{1}{\sqrt{1-4x^2}}$ is $\frac{1}{2}\sin^{-1} 2x$, so

$$\begin{aligned}\int_0^{0.125} \frac{2}{\sqrt{1-4x^2}}\,dx &= 2 \times \tfrac{1}{2}\left[\sin^{-1} 2x\right]_0^{0.125} \\ &= \sin^{-1} 0.25 - \sin^{-1} 0 \\ &= 0.252\,68\ldots \\ &\doteq 0.253\end{aligned}$$

The calculator must be in radians.

PP 7 **Hornsby Girls High School, 2022, Year 12 Higher School Certificate Trial Examination, Mathematics Extension 1, Q11(b)**

Here $a = \sqrt{2}$ in Box 5:

$$\begin{aligned}
\int_{\sqrt{2}}^{\sqrt{6}} \frac{1}{2+x^2}\,dx &= \frac{1}{\sqrt{2}}\left[\tan^{-1}\frac{x}{\sqrt{2}}\right]_{\sqrt{2}}^{\sqrt{6}} \\
&= \frac{1}{\sqrt{2}}\left(\tan^{-1}\frac{\sqrt{6}}{\sqrt{2}} - \tan^{-1}\frac{\sqrt{2}}{\sqrt{2}}\right) \\
&= \frac{1}{\sqrt{2}}\left(\tan^{-1}\sqrt{3} - \tan^{-1}1\right) \\
&= \frac{1}{\sqrt{2}}\left(\frac{\pi}{3} - \frac{\pi}{4}\right) \\
&= \frac{1}{\sqrt{2}} \times \frac{\pi}{12} \\
&= \frac{\sqrt{2}\pi}{24}
\end{aligned}$$

PP 8 **Northern Beaches Secondary College Manly Selective Campus, 2020, Year 12 Trial Examination, Mathematics Extension 1, Q7**

With $a = 2$ in Box 5,

$$\int_0^k \frac{1}{\sqrt{4-x^2}}\,dx = \left[\sin^{-1}\frac{x}{2}\right]_0^k = \sin^{-1}\frac{k}{2} - \sin^{-1}0 = \sin^{-1}\frac{k}{2}.$$

So $\sin^{-1}\dfrac{k}{2} = \dfrac{\pi}{3}$, which gives $\dfrac{k}{2} = \sin\dfrac{\pi}{3} = \dfrac{\sqrt{3}}{2}$ ($\frac{\pi}{3}$ is in the range of $\sin^{-1}$), and $k = \sqrt{3}$. **Answer B.**

PP 9 **James Ruse Agricultural High School, 2023, Year 12 Trial HSC Examination, Mathematics Extension 1, Q13(a)**

With $a = 3$ in Box 5,

$$\begin{aligned}
\left[\sin^{-1}\frac{x}{3}\right]_0^k &= \frac{\pi}{2} \\
\sin^{-1}\frac{k}{3} - \sin^{-1}0 &= \frac{\pi}{2} \\
\sin^{-1}\frac{k}{3} &= \frac{\pi}{2} \\
\frac{k}{3} &= \sin\frac{\pi}{2} = 1 \\
k &= 3
\end{aligned}$$

PP 10 **Girraween High School, 2021, Trial Higher School Certificate Examination, Mathematics Extension 1, Q9**

Let $u = x - 1$, then $u' = 1$. By the reverse chain rule (Box 7), a primitive of $\dfrac{1}{1+(x-1)^2}$ is $\tan^{-1}(x-1)$. So

$$\begin{aligned}
\left[\tan^{-1}(x-1)\right]_0^k &= \frac{\pi}{2} \\
\tan^{-1}(k-1) - \tan^{-1}(-1) &= \frac{\pi}{2} \\
\tan^{-1}(k-1) + \frac{\pi}{4} &= \frac{\pi}{2} \\
\tan^{-1}(k-1) &= \frac{\pi}{4} \\
k - 1 &= 1 \\
k &= 2
\end{aligned}$$

Answer C.

PP 11 **Sydney Technical High School, 2023, Extension 1 Mathematics, Trial HSC, Q14(b)**

The curves meet at $x = \pm 2$: there $y = 4 - \dfrac{8}{\sqrt{8-4}} = 4 - 4 = 0$ and $y = -\sqrt{4-4} = 0$. Split the region by the x-axis into A_1 (above) and A_2 (below).

A_2 is the lower half of the circle $x^2 + y^2 = 4$ of radius 2, so $A_2 = \frac{1}{2} \times \pi \times 2^2 = 2\pi$.

For A_1, use symmetry (the integrand is even) and Box 5 with $a = \sqrt{8} = 2\sqrt{2}$:

$$\begin{aligned}
A_1 &= \int_{-2}^{2} \left(4 - \frac{8}{\sqrt{8-x^2}}\right) dx \\
&= 2\int_{0}^{2} \left(4 - \frac{8}{\sqrt{8-x^2}}\right) dx \\
&= 2\left[4x - 8\sin^{-1}\frac{x}{2\sqrt{2}}\right]_0^2 \\
&= 2\left(\left(8 - 8\sin^{-1}\tfrac{1}{\sqrt{2}}\right) - (0-0)\right) \\
&= 2\left(8 - 8\times\frac{\pi}{4}\right) \\
&= 16 - 4\pi
\end{aligned}$$

So the shaded area is $A_1 + A_2 = 16 - 4\pi + 2\pi = 16 - 2\pi$ square units.

Integration Part 3 Further Integration - Homework Answers

Lesson 12C | Further trigonometric integrals

Lesson Examples

C1 **Tutor demo.**

a

$$\begin{aligned}\int (2\cos x - \sin x)\,dx &= 2\sin x - (-\cos x) + C \\ &= 2\sin x + \cos x + C\end{aligned}$$

b The angle is $3x$, so divide by 3:

$$\int \sec^2 3x\,dx = \tfrac{1}{3}\tan 3x + C$$

c The angle is $\frac{1}{2}x$, so divide by $\frac{1}{2}$, that is, multiply by 2:

$$\begin{aligned}\int_0^{\frac{\pi}{3}} \sin \tfrac{1}{2}x\,dx &= \Big[-2\cos \tfrac{1}{2}x\Big]_0^{\frac{\pi}{3}} \\ &= \big(-2\cos \tfrac{\pi}{6}\big) - \big(-2\cos 0\big) \\ &= -2 \times \tfrac{\sqrt{3}}{2} + 2 \\ &= 2 - \sqrt{3}\end{aligned}$$

Your turn.

a

$$\begin{aligned}\int (3\sin x + \cos x)\,dx &= 3(-\cos x) + \sin x + C \\ &= -3\cos x + \sin x + C\end{aligned}$$

b

$$\int \sec^2 \tfrac{1}{2}x\,dx = 2\tan \tfrac{1}{2}x + C$$

c

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \cos \tfrac{1}{3}x\,dx &= \Big[3\sin \tfrac{1}{3}x\Big]_0^{\frac{\pi}{2}} \\ &= 3\sin \tfrac{\pi}{6} - 3\sin 0 \\ &= 3 \times \tfrac{1}{2} - 0 \\ &= \tfrac{3}{2}\end{aligned}$$

C2 Tutor demo.

a Using $\sin^2 x = \frac{1}{2} - \frac{1}{2}\cos 2x$,

$$\begin{aligned}\int_0^{\frac{\pi}{4}} \sin^2 x\,dx &= \int_0^{\frac{\pi}{4}} \left(\tfrac{1}{2} - \tfrac{1}{2}\cos 2x\right) dx \\ &= \left[\tfrac{1}{2}x - \tfrac{1}{4}\sin 2x\right]_0^{\frac{\pi}{4}} \\ &= \left(\tfrac{\pi}{8} - \tfrac{1}{4}\sin\tfrac{\pi}{2}\right) - \left(0 - \tfrac{1}{4}\sin 0\right) \\ &= \left(\tfrac{\pi}{8} - \tfrac{1}{4}\right) - (0 - 0) \\ &= \tfrac{\pi}{8} - \tfrac{1}{4}\end{aligned}$$

b Using $\cos^2 \frac{1}{2}x = \frac{1}{2} + \frac{1}{2}\cos x$,

$$\begin{aligned}\int_0^{\frac{2\pi}{3}} \cos^2 \tfrac{1}{2}x\,dx &= \int_0^{\frac{2\pi}{3}} \left(\tfrac{1}{2} + \tfrac{1}{2}\cos x\right) dx \\ &= \left[\tfrac{1}{2}x + \tfrac{1}{2}\sin x\right]_0^{\frac{2\pi}{3}} \\ &= \left(\tfrac{\pi}{3} + \tfrac{1}{2}\sin\tfrac{2\pi}{3}\right) - \left(0 + \tfrac{1}{2}\sin 0\right) \\ &= \left(\tfrac{\pi}{3} + \tfrac{1}{2} \times \tfrac{\sqrt{3}}{2}\right) - (0 + 0) \\ &= \tfrac{\pi}{3} + \tfrac{\sqrt{3}}{4}\end{aligned}$$

Your turn.

a Using $\cos^2 x = \frac{1}{2} + \frac{1}{2}\cos 2x$,

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \cos^2 x\,dx &= \int_0^{\frac{\pi}{2}} \left(\tfrac{1}{2} + \tfrac{1}{2}\cos 2x\right) dx \\ &= \left[\tfrac{1}{2}x + \tfrac{1}{4}\sin 2x\right]_0^{\frac{\pi}{2}} \\ &= \left(\tfrac{\pi}{4} + \tfrac{1}{4}\sin\pi\right) - \left(0 + \tfrac{1}{4}\sin 0\right) \\ &= \left(\tfrac{\pi}{4} + 0\right) - (0 + 0) \\ &= \tfrac{\pi}{4}\end{aligned}$$

b Using $\sin^2 \frac{1}{3}x = \frac{1}{2} - \frac{1}{2}\cos\frac{2}{3}x$,

$$\begin{aligned}\int_0^{\pi} \sin^2 \tfrac{1}{3}x\,dx &= \int_0^{\pi} \left(\tfrac{1}{2} - \tfrac{1}{2}\cos\tfrac{2}{3}x\right) dx \\ &= \left[\tfrac{1}{2}x - \tfrac{3}{4}\sin\tfrac{2}{3}x\right]_0^{\pi} \\ &= \left(\tfrac{\pi}{2} - \tfrac{3}{4}\sin\tfrac{2\pi}{3}\right) - \left(0 - \tfrac{3}{4}\sin 0\right) \\ &= \left(\tfrac{\pi}{2} - \tfrac{3}{4} \times \tfrac{\sqrt{3}}{2}\right) - (0 - 0) \\ &= \tfrac{\pi}{2} - \tfrac{3\sqrt{3}}{8}\end{aligned}$$

The primitive of $-\frac{1}{2}\cos\frac{2}{3}x$ is $-\frac{1}{2} \times \frac{3}{2}\sin\frac{2}{3}x = -\frac{3}{4}\sin\frac{2}{3}x$.

C3 **Tutor demo.**

a Using $\tan^2 3x = \sec^2 3x - 1$,

$$\begin{aligned}\int \tan^2 3x \, dx &= \int \left(\sec^2 3x - 1\right) dx \\ &= \tfrac{1}{3}\tan 3x - x + C\end{aligned}$$

b Using $\cot^2 x = \operatorname{cosec}^2 x - 1$,

$$\begin{aligned}\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \cot^2 x \, dx &= \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \left(\operatorname{cosec}^2 x - 1\right) dx \\ &= \Big[-\cot x - x \Big]_{\frac{\pi}{6}}^{\frac{\pi}{4}} \\ &= \left(-\cot \tfrac{\pi}{4} - \tfrac{\pi}{4}\right) - \left(-\cot \tfrac{\pi}{6} - \tfrac{\pi}{6}\right) \\ &= \left(-1 - \tfrac{\pi}{4}\right) - \left(-\sqrt{3} - \tfrac{\pi}{6}\right) \\ &= \sqrt{3} - 1 - \tfrac{\pi}{12}\end{aligned}$$

Your turn.

a Using $\tan^2 \frac{1}{2}x = \sec^2 \frac{1}{2}x - 1$,

$$\begin{aligned}\int \tan^2 \tfrac{1}{2}x \, dx &= \int \left(\sec^2 \tfrac{1}{2}x - 1\right) dx \\ &= 2\tan \tfrac{1}{2}x - x + C\end{aligned}$$

b Using $\tan^2 x = \sec^2 x - 1$,

$$\begin{aligned}\int_0^{\frac{\pi}{3}} \tan^2 x \, dx &= \int_0^{\frac{\pi}{3}} \left(\sec^2 x - 1\right) dx \\ &= \Big[\tan x - x \Big]_0^{\frac{\pi}{3}} \\ &= \left(\tan \tfrac{\pi}{3} - \tfrac{\pi}{3}\right) - \left(\tan 0 - 0\right) \\ &= \sqrt{3} - \tfrac{\pi}{3}\end{aligned}$$

C4 **Tutor demo.**

a Using $2\sin A\cos B = \sin(A+B) + \sin(A-B)$ with $A = 4x$ and $B = x$, $\sin 4x\cos x = \frac{1}{2}(\sin 5x + \sin 3x)$. So

$$\begin{aligned}\int \sin 4x\cos x\,dx &= \frac{1}{2}\int (\sin 5x + \sin 3x)\,dx \\ &= \frac{1}{2}\left(-\frac{1}{5}\cos 5x - \frac{1}{3}\cos 3x\right) + C \\ &= -\frac{1}{10}\cos 5x - \frac{1}{6}\cos 3x + C\end{aligned}$$

b Using $-2\sin A\sin B = \cos(A+B) - \cos(A-B)$ with $A = 3x$ and $B = x$, $2\sin 3x\sin x = -(\cos 4x - \cos 2x) = \cos 2x - \cos 4x$. So

$$\begin{aligned}\int_0^{\frac{\pi}{6}} 2\sin 3x\sin x\,dx &= \int_0^{\frac{\pi}{6}} (\cos 2x - \cos 4x)\,dx \\ &= \left[\frac{1}{2}\sin 2x - \frac{1}{4}\sin 4x\right]_0^{\frac{\pi}{6}} \\ &= \left(\frac{1}{2}\sin\frac{\pi}{3} - \frac{1}{4}\sin\frac{2\pi}{3}\right) - \left(\frac{1}{2}\sin 0 - \frac{1}{4}\sin 0\right) \\ &= \left(\frac{1}{2}\times\frac{\sqrt{3}}{2} - \frac{1}{4}\times\frac{\sqrt{3}}{2}\right) - 0 \\ &= \frac{\sqrt{3}}{4} - \frac{\sqrt{3}}{8} \\ &= \frac{\sqrt{3}}{8}\end{aligned}$$

Your turn.

a Using $2\cos A\sin B = \sin(A+B) - \sin(A-B)$ with $A = 4x$ and $B = 2x$, $\cos 4x\sin 2x = \frac{1}{2}(\sin 6x - \sin 2x)$. So

$$\begin{aligned}\int \cos 4x\sin 2x\,dx &= \frac{1}{2}\int (\sin 6x - \sin 2x)\,dx \\ &= \frac{1}{2}\left(-\frac{1}{6}\cos 6x + \frac{1}{2}\cos 2x\right) + C \\ &= -\frac{1}{12}\cos 6x + \frac{1}{4}\cos 2x + C\end{aligned}$$

b Using $2\cos A\cos B = \cos(A+B) + \cos(A-B)$ with $A = 3x$ and $B = x$, $2\cos 3x\cos x = \cos 4x + \cos 2x$. So

$$\begin{aligned}\int_0^{\frac{\pi}{4}} 2\cos 3x\cos x\,dx &= \int_0^{\frac{\pi}{4}} (\cos 4x + \cos 2x)\,dx \\ &= \left[\frac{1}{4}\sin 4x + \frac{1}{2}\sin 2x\right]_0^{\frac{\pi}{4}} \\ &= \left(\frac{1}{4}\sin\pi + \frac{1}{2}\sin\frac{\pi}{2}\right) - \left(\frac{1}{4}\sin 0 + \frac{1}{2}\sin 0\right) \\ &= \left(0 + \frac{1}{2}\right) - 0 \\ &= \frac{1}{2}\end{aligned}$$

C5 **Tutor demo.**

$$\begin{aligned}\int \tan 2x\,dx &= \int \frac{\sin 2x}{\cos 2x}\,dx \\ &= -\tfrac{1}{2}\int \frac{-2\sin 2x}{\cos 2x}\,dx \\ &= -\tfrac{1}{2}\log_e|\cos 2x| + C\end{aligned}$$

Let $u = \cos 2x$.

Then $u' = -2\sin 2x$,

and $\displaystyle\int \frac{u'}{u}\,dx = \log_e|u|$.

Your turn.

$$\begin{aligned}\int \cot \tfrac{1}{2}x\,dx &= \int \frac{\cos \frac{1}{2}x}{\sin \frac{1}{2}x}\,dx \\ &= 2\int \frac{\frac{1}{2}\cos \frac{1}{2}x}{\sin \frac{1}{2}x}\,dx \\ &= 2\log_e\left|\sin \tfrac{1}{2}x\right| + C\end{aligned}$$

Let $u = \sin \frac{1}{2}x$.

Then $u' = \frac{1}{2}\cos \frac{1}{2}x$,

and $\displaystyle\int \frac{u'}{u}\,dx = \log_e|u|$.

C6 **Tutor demo.**

a

$$\begin{aligned}\int \sin x\cos^6 x\,dx &= -\int(-\sin x)\cos^6 x\,dx \\ &= -\tfrac{1}{7}\cos^7 x + C\end{aligned}$$

Let $u = \cos x$.

Then $\frac{du}{dx} = -\sin x$,

and $\displaystyle\int u^6\frac{du}{dx}\,dx = \tfrac{1}{7}u^7$.

b

$$\int \sec^2 x\tan^n x\,dx = \frac{\tan^{n+1} x}{n+1} + C$$

Let $u = \tan x$.

Then $\frac{du}{dx} = \sec^2 x$,

and $\displaystyle\int u^n\frac{du}{dx}\,dx = \frac{u^{n+1}}{n+1}$.

Your turn.

a

$$\int \cos x\sin^5 x\,dx = \tfrac{1}{6}\sin^6 x + C$$

Let $u = \sin x$.

Then $\frac{du}{dx} = \cos x$,

and $\displaystyle\int u^5\frac{du}{dx}\,dx = \tfrac{1}{6}u^6$.

b

$$\begin{aligned}\int \sin x\cos^n x\,dx &= -\int(-\sin x)\cos^n x\,dx \\ &= -\frac{\cos^{n+1} x}{n+1} + C\end{aligned}$$

Let $u = \cos x$.

Then $\frac{du}{dx} = -\sin x$,

and $\displaystyle\int u^n\frac{du}{dx}\,dx = \frac{u^{n+1}}{n+1}$.

Textbook Homework

2 **a** $\frac{2+\sqrt{3}}{4}$ **b** $\frac{2+\sqrt{3}}{4}$ **c** $-\frac{1}{4}$ **d** $\frac{2-\sqrt{2}}{4}$

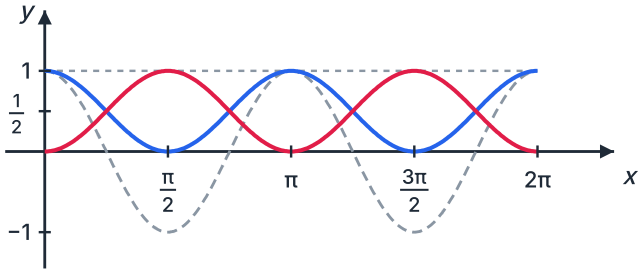
3 **a** $\frac{1}{2}x - \frac{1}{4}\sin 2x + C$ **b** $\frac{1}{2}x - \frac{1}{8}\sin 4x + C$ **c** $\frac{1}{2}x - \sin\frac{1}{2}x + C$
d $\frac{1}{2}x - \frac{1}{12}\sin 6x + C$

4 **a** $\frac{1}{2}x + \frac{1}{4}\sin 2x + C$ **b** $\frac{1}{2}x + \frac{1}{24}\sin 12x + C$ **c** $\frac{1}{2}x + \frac{1}{2}\sin x + C$
d $\frac{1}{2}x + \frac{1}{40}\sin 20x + C$

5 **a** $\frac{\pi}{2}$ **b** $\frac{1}{8}(\pi + 2)$ **c** $\frac{1}{12}(\pi - 3)$ **d** $\frac{1}{32}(\pi + 2\sqrt{2})$ **e** $\frac{1}{24}(4\pi + 9)$
f $\frac{1}{24}(2\pi - 3\sqrt{3})$

6 **a** $-\frac{1}{10}\cos 5x - \frac{1}{2}\cos x + C$ **b** $\frac{1}{4}\cos 2x - \frac{1}{8}\cos 4x + C$ **c** $\frac{2\sqrt{2}}{3}$ **d** $-\frac{\sqrt{3}}{28}$

7 **a, b** $y = \cos 2x$ is dashed, $y = \frac{1}{2}(1 + \cos 2x)$ is blue and $y = \frac{1}{2}(1 - \cos 2x)$ is red.



8 **a** $\frac{1}{4}\sin^4 x + C$ **b** $\frac{1}{7}\sin^7 x + C$ **c** $-\frac{1}{6}\cos^6 x + C$ **d** $-\frac{1}{9}\cos^9 x + C$
e $-\cos e^x + C$ **f** $\frac{1}{5}\sin 5e^x + C$ **g** $-\log_e|\cos x| + C$ **h** $\frac{1}{7}\log_e|\sin 7x| + C$

9 **a** $\cos x \sin x = \frac{1}{2}\sin 2x$, so $-\frac{1}{2} \le y \le \frac{1}{2}$.
b (i) $-\frac{1}{4}\cos 2x + C$
(ii) $\frac{1}{2}\sin^2 x + D$ using $u = \sin x$, and $-\frac{1}{2}\cos^2 x + E$ using $u = \cos x$.
c The three constants C, D and E are different. The answers in part (ii) can be reconciled with each other using the Pythagorean identity, and they can be reconciled with part (i) using the $\cos 2x$ formulae.

10 **b** $\cos^4 x = \frac{3}{8} + \frac{1}{2}\cos 2x + \frac{1}{8}\cos 4x$ **c** (i) $\frac{3\pi}{8}$ (ii) $\frac{1}{32}(3\pi + 8)$ **d** $\frac{5}{24}$

11 **a** $\frac{1}{2}\tan 2x - x + C$ **b** $-2\cot\frac{1}{2}x - x + C$ **c** $\sqrt{3} - 1 - \frac{\pi}{12}$ **d** $\frac{1}{4}\sqrt{3} - \frac{\pi}{12}$

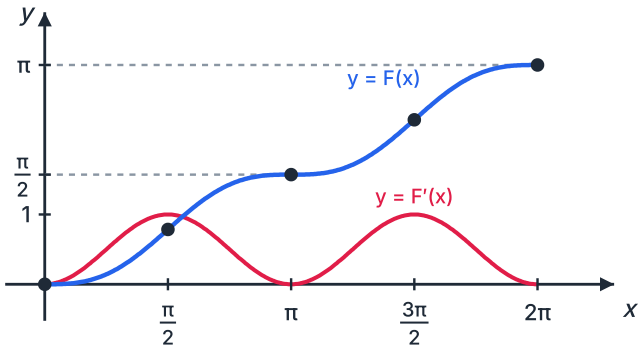
12 **a** $\frac{1}{2}\tan^2 x + C$ **b** $x - \sin x + C$ **c** $\tan x + \sin x + C$

13 **b** (i) $x = 0$, π or 2π (ii) $0 < x < \pi$ or $\pi < x < 2\pi$ (iii) no values of x

c It is because $-\frac{1}{4} \leq \frac{1}{4}\sin 2x \leq \frac{1}{4}$.

d $\left(\frac{\pi}{2}, \frac{\pi}{4}\right)$ and $\left(\frac{3\pi}{2}, \frac{3\pi}{4}\right)$ are points of inflection, and $\left(\pi, \frac{\pi}{2}\right)$ is a stationary (or horizontal) point of inflection.

e $y = F(x)$ is blue and $y = F'(x)$ is red. $F(x)$ is steepest where $F'(x)$ has its maximum, and flat where $F'(x) = 0$.



f (i) $k = 3\pi$ (ii) $k = n\pi$, where n is an integer.

14 $\frac{1}{2}$

Integration Part 3 Further Integration - Homework Answers

Lesson 12C | Further trigonometric integrals

Past-Paper Homework - Worked Solutions

PP 1 **Girraween High School, 2021, Trial Higher School Certificate Examination, Mathematics Extension 1, Q8**

Using $\sin^2\theta = \frac{1}{2} - \frac{1}{2}\cos 2\theta$ with $\theta = 3x$, the angle doubles to $6x$:

$$\begin{aligned}\int \sin^2 3x\,dx &= \int \left(\tfrac{1}{2} - \tfrac{1}{2}\cos 6x\right) dx \\ &= \tfrac{1}{2}x - \tfrac{1}{2} \times \tfrac{1}{6}\sin 6x + C \\ &= \frac{x}{2} - \frac{\sin 6x}{12} + C\end{aligned}$$

Answer (D). Options (A), (B) and (C) divide $\sin 6x$ by 3, 6 and 9 instead of $2 \times 6 = 12$.

PP 2 **Normanhurst Boys High School, 2020, Mathematics Extension 1, Year 12 Course Assessment Task 4 (Trial Examination), Q7**

Using $2\cos A\cos B = \cos(A+B) + \cos(A-B)$ with $A = 3x$ and $B = 2x$, $\cos 3x\cos 2x = \frac{1}{2}(\cos 5x + \cos x)$. So

$$\begin{aligned}\int \cos 3x\ \cos 2x\,dx &= \tfrac{1}{2}\int (\cos 5x + \cos x)\,dx \\ &= \tfrac{1}{2}\left(\tfrac{1}{5}\sin 5x + \sin x\right) + c \\ &= \frac{1}{10}\sin 5x + \frac{1}{2}\sin x + c\end{aligned}$$

Answer (A). Option (B) is the integrand rewritten, not its primitive. Option (C) has cosines where the primitive has sines, and option (D) forgets to divide $\sin 5x$ by 5.

PP 3 **Sydney Grammar School, 2022, Trial Examination, Form VI Mathematics Extension 1, Q11(b)**

Using $\sin^2\theta = \frac{1}{2}(1 - \cos 2\theta)$ with $\theta = 5x$,

$$\begin{aligned}\int \sin^2 5x\,dx &= \int \tfrac{1}{2}(1 - \cos 10x)\,dx \\ &= \tfrac{1}{2}\left(x - \tfrac{1}{10}\sin 10x\right) + C \\ &= \frac{x}{2} - \frac{\sin 10x}{20} + C\end{aligned}$$

PP 4 **Caringbah High School, 2021, Year 12 Mathematics Extension 1 HSC Course Assessment Task 4, Q11(a)**

Using $\sin^2\theta = \frac{1}{2} - \frac{1}{2}\cos 2\theta$ with $\theta = 2x$,

$$\begin{aligned}\int_0^{\frac{\pi}{6}} \sin^2 2x\,dx &= \int_0^{\frac{\pi}{6}} \left(\tfrac{1}{2} - \tfrac{1}{2}\cos 4x\right) dx \\ &= \left[\tfrac{1}{2}x - \tfrac{1}{8}\sin 4x\right]_0^{\frac{\pi}{6}} \\ &= \left(\tfrac{1}{2}\times\tfrac{\pi}{6} - \tfrac{1}{8}\sin\tfrac{2\pi}{3}\right) - \left(0 - \tfrac{1}{8}\sin 0\right) \\ &= \tfrac{\pi}{12} - \tfrac{1}{8}\times\tfrac{\sqrt{3}}{2} - 0 \\ &= \tfrac{\pi}{12} - \tfrac{\sqrt{3}}{16}\end{aligned}$$

PP 5 **Hornsby Girls High School, 2022, Mathematics Extension 1, Year 12 Higher School Certificate Trial Examination Term 3, Q11(c)**

A square is never negative, so $y = \cos^2(3x) \geq 0$ and the area is the integral. Using $\cos^2\theta = \frac{1}{2} + \frac{1}{2}\cos 2\theta$ with $\theta = 3x$,

$$\begin{aligned}A &= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \cos^2(3x)\,dx \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \left(\tfrac{1}{2} + \tfrac{1}{2}\cos 6x\right) dx \\ &= \left[\tfrac{1}{2}x + \tfrac{1}{12}\sin 6x\right]_{\frac{\pi}{6}}^{\frac{\pi}{3}} \\ &= \left(\tfrac{\pi}{6} + \tfrac{1}{12}\sin 2\pi\right) - \left(\tfrac{\pi}{12} + \tfrac{1}{12}\sin\pi\right) \\ &= \left(\tfrac{\pi}{6} + 0\right) - \left(\tfrac{\pi}{12} + 0\right) \\ &= \tfrac{\pi}{12}\end{aligned}$$

The area is $\frac{\pi}{12}$ square units.

PP 6 **North Sydney Girls High School, 2023, HSC Trial Examination, Mathematics Extension 1, Q12(a)**

Expand the square, then use $\cos^2\theta = \frac{1}{2}(1 + \cos 2\theta)$:

$$\begin{aligned}\int \left(1 + \cos\theta\right)^2 d\theta &= \int \left(1 + 2\cos\theta + \cos^2\theta\right) d\theta \\ &= \int \left(1 + 2\cos\theta + \tfrac{1}{2}\left(1 + \cos 2\theta\right)\right) d\theta \\ &= \int \left(\tfrac{3}{2} + 2\cos\theta + \tfrac{1}{2}\cos 2\theta\right) d\theta \\ &= \tfrac{3}{2}\theta + 2\sin\theta + \tfrac{1}{4}\sin 2\theta + C\end{aligned}$$

PP 7 **Cherrybrook Technology High School, 2020, Year 12 AP4, Mathematics Extension 1, Q12(a)**

(i) Using the compound-angle formulae for $\sin(A+B)$ and $\sin(A-B)$,

$$\begin{aligned}\text{LHS} &= \big(\sin A\cos B + \cos A\sin B\big) + \big(\sin A\cos B - \cos A\sin B\big)\\ &= 2\sin A\cos B\\ &= \text{RHS}.\end{aligned}$$

(ii) By part (i) with $A = 4x$ and $B = 2x$, $\sin 4x\cos 2x = \frac{1}{2}\big(\sin 6x + \sin 2x\big)$. So

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \sin 4x\cos 2x\,dx &= \int_0^{\frac{\pi}{2}} \tfrac{1}{2}\big(\sin 6x + \sin 2x\big)\,dx\\ &= \tfrac{1}{2}\Big[-\tfrac{1}{6}\cos 6x - \tfrac{1}{2}\cos 2x\Big]_0^{\frac{\pi}{2}}\\ &= \tfrac{1}{2}\Big(\big(-\tfrac{1}{6}\cos 3\pi - \tfrac{1}{2}\cos\pi\big) - \big(-\tfrac{1}{6}\cos 0 - \tfrac{1}{2}\cos 0\big)\Big)\\ &= \tfrac{1}{2}\Big(\big(\tfrac{1}{6}+\tfrac{1}{2}\big) - \big(-\tfrac{1}{6}-\tfrac{1}{2}\big)\Big)\\ &= \tfrac{1}{2}\times\tfrac{4}{3}\\ &= \tfrac{2}{3}\end{aligned}$$

PP 8 **Ascham School, 2020, Mathematics Extension 1 Trial HSC Examination, Q14(b)**

Write $\cos^4 x = (\cos^2 x)^2$ and use $\cos^2\theta = \frac{1}{2} + \frac{1}{2}\cos 2\theta$ twice, first with $\theta = x$, then with $\theta = 2x$:

$$\begin{aligned}\cos^4 x &= \big(\tfrac{1}{2} + \tfrac{1}{2}\cos 2x\big)^2\\ &= \tfrac{1}{4} + \tfrac{1}{2}\cos 2x + \tfrac{1}{4}\cos^2 2x\\ &= \tfrac{1}{4} + \tfrac{1}{2}\cos 2x + \tfrac{1}{4}\big(\tfrac{1}{2} + \tfrac{1}{2}\cos 4x\big)\\ &= \tfrac{3}{8} + \tfrac{1}{2}\cos 2x + \tfrac{1}{8}\cos 4x\end{aligned}$$

So

$$\begin{aligned}\int_0^{\frac{\pi}{8}} \cos^4 x\,dx &= \int_0^{\frac{\pi}{8}} \big(\tfrac{3}{8} + \tfrac{1}{2}\cos 2x + \tfrac{1}{8}\cos 4x\big)\,dx\\ &= \Big[\tfrac{3}{8}x + \tfrac{1}{4}\sin 2x + \tfrac{1}{32}\sin 4x\Big]_0^{\frac{\pi}{8}}\\ &= \big(\tfrac{3\pi}{64} + \tfrac{1}{4}\sin\tfrac{\pi}{4} + \tfrac{1}{32}\sin\tfrac{\pi}{2}\big) - \big(0 + \tfrac{1}{4}\sin 0 + \tfrac{1}{32}\sin 0\big)\\ &= \tfrac{3\pi}{64} + \tfrac{1}{4}\times\tfrac{1}{\sqrt{2}} + \tfrac{1}{32}\\ &= \tfrac{3\pi}{64} + \tfrac{\sqrt{2}}{8} + \tfrac{1}{32}\\ &= \frac{3\pi + 8\sqrt{2} + 2}{64}\end{aligned}$$

Integration Part 3 Further Integration - Homework Answers

Lesson 12D | Integration by substitution

Lesson Examples

C1 **Tutor demo.**

$$\begin{aligned}\int x(3-x^2)^5\,dx &= -\tfrac{1}{2}\int(-2x)(3-x^2)^5\,dx \\ &= -\tfrac{1}{2}\times\tfrac{1}{6}(3-x^2)^6+C \\ &= -\tfrac{1}{12}(3-x^2)^6+C\end{aligned}$$

Let $u=3-x^2$.

Then $\dfrac{du}{dx}=-2x$,

and $\displaystyle\int u^5\,\frac{du}{dx}\,dx=\tfrac{1}{6}u^6$.

Your turn.

$$\begin{aligned}\int x(5-x^2)^3\,dx &= -\tfrac{1}{2}\int(-2x)(5-x^2)^3\,dx \\ &= -\tfrac{1}{2}\times\tfrac{1}{4}(5-x^2)^4+C \\ &= -\tfrac{1}{8}(5-x^2)^4+C\end{aligned}$$

Let $u=5-x^2$.

Then $\dfrac{du}{dx}=-2x$,

and $\displaystyle\int u^3\,\frac{du}{dx}\,dx=\tfrac{1}{4}u^4$.

C2 Tutor demo.

a

$$\begin{aligned}\int x(3-x^2)^5\,dx &= \int u^5\left(-\tfrac{1}{2}\right)du \\ &= -\tfrac{1}{2}\times\tfrac{1}{6}u^6+C \\ &= -\tfrac{1}{12}(3-x^2)^6+C\end{aligned}$$

Let $u=3-x^2$.

Then $du=-2x\,dx$,

and $x\,dx=-\tfrac{1}{2}\,du$.

This is the same answer as the Concept 1 demo.

b

$$\begin{aligned}\int \sin x\sqrt{1+\cos x}\,dx &= \int u^{\frac{1}{2}}(-du) \\ &= -\tfrac{2}{3}u^{\frac{3}{2}}+C \\ &= -\tfrac{2}{3}(1+\cos x)^{\frac{3}{2}}+C\end{aligned}$$

Let $u=1+\cos x$.

Then $du=-\sin x\,dx$,

and $\sin x\,dx=-du$.

Your turn.

a

$$\begin{aligned}\int x(5-x^2)^3\,dx &= \int u^3\left(-\tfrac{1}{2}\right)du \\ &= -\tfrac{1}{2}\times\tfrac{1}{4}u^4+C \\ &= -\tfrac{1}{8}(5-x^2)^4+C\end{aligned}$$

Let $u=5-x^2$.

Then $du=-2x\,dx$,

and $x\,dx=-\tfrac{1}{2}\,du$.

b

$$\begin{aligned}\int \cos x\sqrt{3+\sin x}\,dx &= \int u^{\frac{1}{2}}\,du \\ &= \tfrac{2}{3}u^{\frac{3}{2}}+C \\ &= \tfrac{2}{3}(3+\sin x)^{\frac{3}{2}}+C\end{aligned}$$

Let $u=3+\sin x$.

Then $du=\cos x\,dx$.

C3 **Tutor demo.**

$$\begin{aligned}\int x\sqrt{2-x}\,dx &= \int (2-u)\sqrt{u}\,(-du) \\ &= \int (u-2)\sqrt{u}\,du \\ &= \int \left(u^{\frac{3}{2}} - 2u^{\frac{1}{2}}\right) du \\ &= \tfrac{2}{5}u^{\frac{5}{2}} - 2\times\tfrac{2}{3}u^{\frac{3}{2}} + C \\ &= \tfrac{2}{5}(2-x)^{\frac{5}{2}} - \tfrac{4}{3}(2-x)^{\frac{3}{2}} + C\end{aligned}$$

Let $u = 2 - x$.

Then $du = -dx$,

and $x = 2 - u$.

Your turn.

$$\begin{aligned}\int x\sqrt{x+3}\,dx &= \int (u-3)\sqrt{u}\,du \\ &= \int \left(u^{\frac{3}{2}} - 3u^{\frac{1}{2}}\right) du \\ &= \tfrac{2}{5}u^{\frac{5}{2}} - 3\times\tfrac{2}{3}u^{\frac{3}{2}} + C \\ &= \tfrac{2}{5}(x+3)^{\frac{5}{2}} - 2(x+3)^{\frac{3}{2}} + C\end{aligned}$$

Let $u = x + 3$.

Then $du = dx$,

and $x = u - 3$.

C4 Tutor demo.

a

$$\begin{aligned}\int_0^2 x\sqrt{2-x}\,dx &= -\int_2^0 (2-u)\sqrt{u}\,du \\ &= -\int_2^0 \left(2u^{\frac{1}{2}} - u^{\frac{3}{2}}\right) du \\ &= -\left[\tfrac{4}{3}u^{\frac{3}{2}} - \tfrac{2}{5}u^{\frac{5}{2}}\right]_2^0 \\ &= -\left(0-0\right) + \left(\tfrac{4}{3}\times 2\sqrt{2} - \tfrac{2}{5}\times 4\sqrt{2}\right) \\ &= \tfrac{8\sqrt{2}}{3} - \tfrac{8\sqrt{2}}{5} \\ &= \tfrac{40\sqrt{2}-24\sqrt{2}}{15} \\ &= \tfrac{16\sqrt{2}}{15}\end{aligned}$$

Let $u = 2-x$.

Then $du = -dx$,

and $x = 2-u$.

When $x=0$, $u=2$,

when $x=2$, $u=0$.

The primitive of $2u^{\frac{1}{2}}$ is $2\times\frac{2}{3}u^{\frac{3}{2}} = \frac{4}{3}u^{\frac{3}{2}}$. Also $2^{\frac{3}{2}} = 2\sqrt{2}$ and $2^{\frac{5}{2}} = 2^2\times\sqrt{2} = 4\sqrt{2}$.

b

$$\begin{aligned}\int_0^{\pi} \sin x\cos^4 x\,dx &= -\int_1^{-1} u^4\,du \\ &= -\tfrac{1}{5}\left[u^5\right]_1^{-1} \\ &= -\tfrac{1}{5}\left((-1)^5 - 1^5\right) \\ &= -\tfrac{1}{5}\times(-2) \\ &= \tfrac{2}{5}\end{aligned}$$

Let $u = \cos x$.

Then $du = -\sin x\,dx$.

When $x=0$, $u=1$,

when $x=\pi$, $u=-1$.

Your turn.

a

$$\begin{aligned}\int_{-3}^{1} x\sqrt{x+3}\,dx &= \int_0^4 (u-3)\sqrt{u}\,du \\ &= \int_0^4 \left(u^{\frac{3}{2}} - 3u^{\frac{1}{2}}\right) du \\ &= \left[\tfrac{2}{5}u^{\frac{5}{2}} - 2u^{\frac{3}{2}}\right]_0^4 \\ &= \left(\tfrac{2}{5}\times 32 - 2\times 8\right) - \left(0-0\right) \\ &= \tfrac{64}{5} - 16 \\ &= -\tfrac{16}{5}\end{aligned}$$

Let $u = x+3$.

Then $du = dx$,

and $x = u-3$.

When $x=-3$, $u=0$,

when $x=1$, $u=4$.

$4^{\frac{5}{2}} = \left(\sqrt{4}\right)^5 = 2^5 = 32$ and $4^{\frac{3}{2}} = 2^3 = 8$. The value is negative because the integrand is negative for $-3 < x < 0$.

b

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \sin x \cos^5 x \, dx &= -\int_1^0 u^5 \, du \\ &= -\tfrac{1}{6}\left[u^6\right]_1^0 \\ &= -\tfrac{1}{6}\left(0^6 - 1^6\right) \\ &= \tfrac{1}{6}\end{aligned}$$

Let $u = \cos x$.

Then $du = -\sin x \, dx$.

When $x = 0$, $u = 1$,

when $x = \frac{\pi}{2}$, $u = 0$.

Textbook Homework

1 **c** $\frac{1}{4}(1+x^2)^4 + C$

2 **a** $\frac{1}{4}(2x+3)^4 + C$ **b** $\frac{1}{5}(1+x^3)^5 + C$ **c** $\dfrac{-1}{1+x^2} + C$ **d** $2\sqrt{3x-5} + C$
e $\frac{1}{4}\sin^4 x + C$ **f** $\log_e(1+x^4) + C$

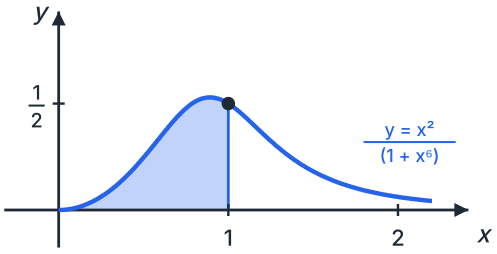
3 **c** $-\sqrt{1-x^2} + C$

4 **a** $\frac{1}{24}(x^4+1)^6 + C$ **b** $\frac{2}{9}(x^3-1)^{\frac{3}{2}} + C$ **c** $\frac{1}{3}e^{x^3} + C$
d $\dfrac{-1}{(1+\sqrt{x})^2} + C$ **e** $\frac{1}{6}\tan^3 2x + C$ **f** $-e^{\frac{1}{x}} + C$

5 **a** $\frac{65}{12}$ **b** $\sqrt{2} - 1$ **c** $\frac{1}{3}$ **d** $\frac{1}{24}$ **e** 2 **f** $\frac{1}{2}(e^2 - 1)$ **g** $\frac{1}{10}$ **h** $\frac{\pi^4}{64}$ **i** 3 **j** $\frac{1}{2}\ln 3$

6 $\frac{\pi}{12}$ units^2

7 **a** $\log_e \frac{3}{2}$ **b** $\frac{\pi}{4}$ **c** $\frac{2}{3}$ **d** $\frac{4}{3}$



8 **a** $\sqrt{1+e^{2x}} + C$ **b** $\ln|\ln x| + C$ **c** $-\ln|\ln(\cos x)| + C$
d $\frac{1}{4}\tan^4 x + \frac{1}{6}\tan^6 x + C$

9 **a** $y = \frac{1}{2}\tan^{-1} e^{2x}$
b $y' = \dfrac{1}{\sqrt{4-x^2}} + \frac{1}{2}$ and $y = \sin^{-1}\dfrac{x}{2} + \dfrac{x}{2} + \dfrac{1}{2}$

10 **b** (i) $\dfrac{2}{\ln 2}$ (ii) $\frac{1}{5}(4\sqrt{2} - 1)$

11 **a** $\ln 2$ **b** $\dfrac{e}{e+1}$

12 $\tan^{-1}\sqrt{x-1} + C$

13 **a** $2\sin^{-1}\sqrt{x} + C_1$ **b** $\sin^{-1}(2x-1) + C_2$

Integration Part 3 Further Integration - Homework Answers

Lesson 12D | Integration by substitution

Past-Paper Homework - Worked Solutions

PP 1 **Blacktown Boys' High School, 2020, HSC Trial Examination, Mathematics Extension 1, Q8**

$$\int_0^{\frac{\pi}{3}} \tan^2 x \sec^2 x \, dx = \int_0^{\sqrt{3}} u^2 \, du$$

Let $u = \tan x$.

Then $du = \sec^2 x \, dx$.

When $x = 0$, $u = 0$,

when $x = \frac{\pi}{3}$, $u = \sqrt{3}$.

Answer (B). Option (A) keeps the x-limits on a u-integral. Option (C) loses the square on $\tan x$. Option (D) writes $\sec^2 x$ as $1 + u^2$ as well, but $\sec^2 x \, dx$ has already become du.

PP 2 **Moriah College, 2023, Trial Higher School Certificate Examination, Mathematics Extension 1, Q10(a)**

$$\begin{aligned}\int x\sqrt{x^2+1}\, dx &= \int u^{\frac{1}{2}} \times \tfrac{1}{2}\, du \\ &= \tfrac{1}{2} \times \tfrac{2}{3} u^{\frac{3}{2}} + C \\ &= \tfrac{1}{3}(x^2+1)^{\frac{3}{2}} + C\end{aligned}$$

Let $u = x^2 + 1$.

Then $du = 2x \, dx$,

and $x \, dx = \frac{1}{2} \, du$.

PP 3 **Sydney Grammar School, 2020, Trial Examination, Form VI Mathematics Extension 1, Q11(d)**

From $u = 2x - 1$, $2x = u + 1$, so $4x = 2(u+1)$.

$$\begin{aligned}\int 4x\sqrt{2x-1}\, dx &= \int 2(u+1)\sqrt{u} \times \tfrac{1}{2}\, du \\ &= \int (u+1)\sqrt{u}\, du \\ &= \int \left(u^{\frac{3}{2}} + u^{\frac{1}{2}}\right) du \\ &= \tfrac{2}{5}u^{\frac{5}{2}} + \tfrac{2}{3}u^{\frac{3}{2}} + C \\ &= \tfrac{2}{5}(2x-1)^{\frac{5}{2}} + \tfrac{2}{3}(2x-1)^{\frac{3}{2}} + C\end{aligned}$$

Let $u = 2x - 1$.

Then $du = 2 \, dx$,

and $dx = \frac{1}{2} \, du$.

PP 4 **Girraween High School, 2021, Trial Higher School Certificate Examination, Mathematics Extension 1, Q11(c)**

By the chain rule, $\dfrac{d}{dx}\ln 3x = \dfrac{3}{3x} = \dfrac{1}{x}$.

$$\begin{aligned}\int \frac{dx}{x(\ln 3x)^2} &= \int \frac{1}{u^2}\,du \\ &= \int u^{-2}\,du \\ &= -u^{-1} + C \\ &= -\frac{1}{\ln 3x} + C\end{aligned}$$

Let $u = \ln 3x$.

Then $du = \dfrac{1}{x}\,dx$.

PP 5 **Girraween High School, 2023, Year 12 Trial Examination, Mathematics Extension 1, Q11(b)**

$$\begin{aligned}\int_1^6 x\sqrt{6-x}\,dx &= -\int_5^0 (6-u)\sqrt{u}\,du \\ &= \int_0^5 (6-u)\sqrt{u}\,du \\ &= \int_0^5 \left(6u^{\frac{1}{2}} - u^{\frac{3}{2}}\right) du \\ &= \left[4u^{\frac{3}{2}} - \tfrac{2}{5}u^{\frac{5}{2}}\right]_0^5 \\ &= \left(4\times 5\sqrt{5} - \tfrac{2}{5}\times 25\sqrt{5}\right) - (0-0) \\ &= 20\sqrt{5} - 10\sqrt{5} \\ &= 10\sqrt{5}\end{aligned}$$

Let $u = 6 - x$.

Then $du = -dx$,

and $x = 6 - u$.

When $x = 1$, $u = 5$,

when $x = 6$, $u = 0$.

The second line reverses the limits to remove the minus sign. The primitive of $6u^{\frac{1}{2}}$ is $6 \times \frac{2}{3}u^{\frac{3}{2}} = 4u^{\frac{3}{2}}$. Also $5^{\frac{3}{2}} = 5\sqrt{5}$ and $5^{\frac{5}{2}} = 25\sqrt{5}$.

PP 6 **Abbotsleigh, 2023, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q12(a)**

After the substitution, use $\cos^2 u = \frac{1}{2} + \frac{1}{2}\cos 2u$ (Lesson 12C).

$$\begin{aligned}\int \frac{\cos^2(\ln x)}{x}\,dx &= \int \cos^2 u\,du \\ &= \int \left(\tfrac{1}{2} + \tfrac{1}{2}\cos 2u\right) du \\ &= \tfrac{1}{2}u + \tfrac{1}{4}\sin 2u + C \\ &= \tfrac{1}{2}\ln x + \tfrac{1}{4}\sin(2\ln x) + C\end{aligned}$$

Let $u = \ln x$.

Then $du = \dfrac{1}{x}\,dx$.

PP 7 **Knox Grammar School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q11(d)**

The derivative of $\tan x - x$ is $\sec^2 x - 1$, and $\sec^2 x - 1 = \tan^2 x$ (Lesson 12C). So $du = \tan^2 x \, dx$.

$$\begin{aligned}\int \tan^2 x \sqrt{\tan x - x} \, dx &= \int u^{\frac{1}{2}} \, du \\ &= \tfrac{2}{3} u^{\frac{3}{2}} + C \\ &= \tfrac{2}{3} (\tan x - x)^{\frac{3}{2}} + C\end{aligned}$$

Let $u = \tan x - x$.

Then $du = \tan^2 x \, dx$.

PP 8 **Sydney Technical High School, 2021, Extension 1 Mathematics, Trial HSC, Q12(b)**

Since $4e^{2x} = \left(2e^x\right)^2 = u^2$, and $e^x \, dx = \frac{1}{2} \, du$:

$$\begin{aligned}\int_{\ln\left(\frac{1}{2}\right)}^{\ln\left(\frac{\sqrt{3}}{2}\right)} \frac{e^x}{1 + 4e^{2x}} \, dx &= \tfrac{1}{2} \int_1^{\sqrt{3}} \frac{du}{1 + u^2} \\ &= \tfrac{1}{2} \left[\tan^{-1} u \right]_1^{\sqrt{3}} \\ &= \tfrac{1}{2} \left(\tan^{-1} \sqrt{3} - \tan^{-1} 1 \right) \\ &= \tfrac{1}{2} \left(\tfrac{\pi}{3} - \tfrac{\pi}{4} \right) \\ &= \tfrac{1}{2} \times \tfrac{\pi}{12} \\ &= \tfrac{\pi}{24}\end{aligned}$$

Let $u = 2e^x$.

Then $du = 2e^x \, dx$,

and $e^x \, dx = \frac{1}{2} \, du$.

When $x = \ln \frac{1}{2}$, $u = 1$,

when $x = \ln \frac{\sqrt{3}}{2}$, $u = \sqrt{3}$.

$u = 2e^{\ln \frac{1}{2}} = 2 \times \frac{1}{2} = 1$ and $u = 2e^{\ln \frac{\sqrt{3}}{2}} = 2 \times \frac{\sqrt{3}}{2} = \sqrt{3}$.

Integration Part 3 Further Integration - Homework Answers

Lesson 12E | Further integration by substitution

Lesson Examples

C1 **Tutor demo.**

a

$$\begin{aligned}\int x\sqrt{x-2}\,dx &= \int (u^2+2)\,u \times 2u\,du \\ &= \int (2u^4+4u^2)\,du \\ &= \tfrac{2}{5}u^5 + \tfrac{4}{3}u^3 + C \\ &= \tfrac{2}{5}(x-2)^{\frac{5}{2}} + \tfrac{4}{3}(x-2)^{\frac{3}{2}} + C\end{aligned}$$

Let $x = u^2+2$.

Then $dx = 2u\,du$,

and $\sqrt{x-2} = u$.

Here $x-2 = u^2$ and $u \geq 0$, so $u = \sqrt{x-2}$. Then $u^5 = (x-2)^{\frac{5}{2}}$ and $u^3 = (x-2)^{\frac{3}{2}}$.

b

$$\begin{aligned}\int_{-3}^{1} x\sqrt{1-x}\,dx &= \int_{2}^{0} (1-u^2)\,u\,(-2u)\,du \\ &= -2\int_{2}^{0} (u^2-u^4)\,du \\ &= -2\left[\tfrac{1}{3}u^3 - \tfrac{1}{5}u^5\right]_{2}^{0} \\ &= -2\left(0 - \left(\tfrac{8}{3} - \tfrac{32}{5}\right)\right) \\ &= 2\left(\tfrac{8}{3} - \tfrac{32}{5}\right) \\ &= 2 \times \left(-\tfrac{56}{15}\right) \\ &= -\tfrac{112}{15}\end{aligned}$$

Let $x = 1-u^2$.

Then $dx = -2u\,du$,

and $\sqrt{1-x} = u$.

When $x = -3$, $u = 2$,

when $x = 1$, $u = 0$.

When $x = -3$, $1-u^2 = -3$, so $u^2 = 4$ and $u = 2$ (taking $u \geq 0$). The value is negative because $x\sqrt{1-x} < 0$ for $-3 < x < 0$, and that part outweighs the part from 0 to 1.

Your turn.

a

$$\begin{aligned}\int x\sqrt{x+3}\,dx &= \int (u^2-3)\,u \times 2u\,du \\ &= \int (2u^4 - 6u^2)\,du \\ &= \tfrac{2}{5}u^5 - 2u^3 + C \\ &= \tfrac{2}{5}(x+3)^{\frac{5}{2}} - 2(x+3)^{\frac{3}{2}} + C\end{aligned}$$

Let $x = u^2 - 3$.

Then $dx = 2u\,du$,

and $\sqrt{x+3} = u$.

b

$$\begin{aligned}\int_{-3}^{0} x\sqrt{1-x}\,dx &= \int_{2}^{1} (1-u^2)\,u\,(-2u)\,du \\ &= -2\int_{2}^{1} (u^2 - u^4)\,du \\ &= -2\left[\tfrac{1}{3}u^3 - \tfrac{1}{5}u^5\right]_{2}^{1} \\ &= -2\left(\left(\tfrac{1}{3} - \tfrac{1}{5}\right) - \left(\tfrac{8}{3} - \tfrac{32}{5}\right)\right) \\ &= -2\left(\tfrac{2}{15} + \tfrac{56}{15}\right) \\ &= -2 \times \tfrac{58}{15} \\ &= -\tfrac{116}{15}\end{aligned}$$

Let $x = 1 - u^2$.

Then $dx = -2u\,du$,

and $\sqrt{1-x} = u$.

When $x = -3$, $u = 2$,

when $x = 0$, $u = 1$.

C2 Tutor demo.

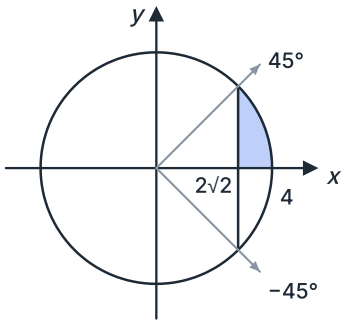
a

Let $x = 4\sin u$. Then $dx = 4\cos u\,du$, and $\sqrt{16 - x^2} = 4\cos u$. When $x = 2\sqrt{2}$, $u = \frac{\pi}{4}$, when $x = 4$, $u = \frac{\pi}{2}$.

$$\begin{aligned}
\int_{2\sqrt{2}}^{4} \sqrt{16 - x^2}\,dx &= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} 4\cos u \times 4\cos u\,du \\
&= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} 16\cos^2 u\,du \\
&= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} 16\left(\tfrac{1}{2} + \tfrac{1}{2}\cos 2u\right) du \\
&= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (8 + 8\cos 2u)\,du \\
&= \Big[8u + 4\sin 2u\Big]_{\frac{\pi}{4}}^{\frac{\pi}{2}} \\
&= \left(4\pi + 4\sin\pi\right) - \left(2\pi + 4\sin\tfrac{\pi}{2}\right) \\
&= (4\pi + 0) - (2\pi + 4) \\
&= 2\pi - 4
\end{aligned}$$

Here $16 - x^2 = 16 - 16\sin^2 u = 16\cos^2 u$, and $\cos u \geq 0$ for $\frac{\pi}{4} \leq u \leq \frac{\pi}{2}$. When $x = 2\sqrt{2}$, $\sin u = \frac{2\sqrt{2}}{4} = \frac{1}{\sqrt{2}}$.

b The chord $x = 2\sqrt{2}$ meets the circle $x^2 + y^2 = 16$ at the angles $45°$ and $-45°$, so it subtends $90° = \frac{\pi}{2}$ at the centre. The region is half of that segment:



$$\begin{aligned}
\int_{2\sqrt{2}}^{4} \sqrt{16 - x^2}\,dx &= \tfrac{1}{2} \times \tfrac{1}{2} \times 4^2\left(\tfrac{\pi}{2} - \sin\tfrac{\pi}{2}\right) \\
&= 4\left(\tfrac{\pi}{2} - 1\right) \\
&= 2\pi - 4
\end{aligned}$$

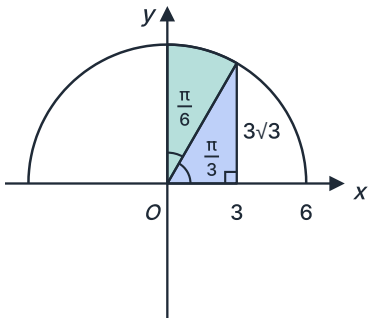
Your turn.

a

Let $x = 6\sin u$. Then $dx = 6\cos u\,du$, and $\sqrt{36 - x^2} = 6\cos u$. When $x = 0$, $u = 0$, when $x = 3$, $u = \frac{\pi}{6}$.

$$\begin{aligned}\int_0^3 \sqrt{36 - x^2}\,dx &= \int_0^{\frac{\pi}{6}} 6\cos u \times 6\cos u\,du \\ &= \int_0^{\frac{\pi}{6}} 36\cos^2 u\,du \\ &= \int_0^{\frac{\pi}{6}} 36\left(\tfrac{1}{2} + \tfrac{1}{2}\cos 2u\right)du \\ &= \int_0^{\frac{\pi}{6}} (18 + 18\cos 2u)\,du \\ &= \left[18u + 9\sin 2u\right]_0^{\frac{\pi}{6}} \\ &= \left(3\pi + 9\sin\tfrac{\pi}{3}\right) - (0 + 9\sin 0) \\ &= 3\pi + 9 \times \tfrac{\sqrt{3}}{2} \\ &= 3\pi + \tfrac{9\sqrt{3}}{2}\end{aligned}$$

b At $x = 3$, $y = \sqrt{36 - 9} = 3\sqrt{3}$, and the radius to $(3, 3\sqrt{3})$ makes the angle $\frac{\pi}{3}$ with the x-axis, because $\cos\frac{\pi}{3} = \frac{3}{6}$. The region is the triangle with vertices O, $(3, 0)$ and $(3, 3\sqrt{3})$, together with the sector between that radius and the y-axis. The angle of the sector is $\frac{\pi}{2} - \frac{\pi}{3} = \frac{\pi}{6}$. So



$$\begin{aligned}\int_0^3 \sqrt{36 - x^2}\,dx &= \tfrac{1}{2} \times 3 \times 3\sqrt{3} + \tfrac{1}{2} \times 6^2 \times \tfrac{\pi}{6} \\ &= \tfrac{9\sqrt{3}}{2} + 3\pi\end{aligned}$$

C3 Tutor demo.

a Take $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$, so $\sec\theta > 0$.

$$\begin{aligned}\int \frac{1}{(9+x^2)^{\frac{3}{2}}}\,dx &= \int \frac{3\sec^2\theta}{27\sec^3\theta}\,d\theta \\ &= \tfrac{1}{9}\int \cos\theta\,d\theta \\ &= \tfrac{1}{9}\sin\theta + C \\ &= \frac{x}{9\sqrt{9+x^2}} + C\end{aligned}$$

Let $x = 3\tan\theta$.

Then $dx = 3\sec^2\theta\,d\theta$,

and $(9+x^2)^{\frac{3}{2}} = 27\sec^3\theta$.

Here $9 + x^2 = 9 + 9\tan^2\theta = 9\sec^2\theta$, so $(9+x^2)^{\frac{3}{2}} = (3\sec\theta)^3 = 27\sec^3\theta$. To return to x: $\tan\theta = \frac{x}{3}$, so $\sec\theta = \sqrt{1+\tan^2\theta} = \frac{\sqrt{9+x^2}}{3}$ and $\sin\theta = \frac{\tan\theta}{\sec\theta} = \frac{x}{\sqrt{9+x^2}}$.

b Take $0 \le \theta \le \pi$, so $\sin\theta \ge 0$.

$$\begin{aligned}\int_0^1 \frac{1}{\sqrt{4-x^2}}\,dx &= \int_{\frac{\pi}{2}}^{\frac{\pi}{3}} \frac{-2\sin\theta}{2\sin\theta}\,d\theta \\ &= \int_{\frac{\pi}{2}}^{\frac{\pi}{3}} (-1)\,d\theta \\ &= \Big[-\theta\Big]_{\frac{\pi}{2}}^{\frac{\pi}{3}} \\ &= -\tfrac{\pi}{3} + \tfrac{\pi}{2} \\ &= \tfrac{\pi}{6}\end{aligned}$$

Let $x = 2\cos\theta$.

Then $dx = -2\sin\theta\,d\theta$,

and $\sqrt{4-x^2} = 2\sin\theta$.

When $x = 0$, $\theta = \frac{\pi}{2}$,

when $x = 1$, $\theta = \frac{\pi}{3}$.

Here $4 - x^2 = 4 - 4\cos^2\theta = 4\sin^2\theta$. When $x = 1$, $\cos\theta = \frac{1}{2}$. The answer agrees with $\sin^{-1}\frac{1}{2} - \sin^{-1}0 = \frac{\pi}{6}$ from Lesson 12B.

Your turn.

a Take $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$, so $\cos\theta \geq 0$.

$$\begin{aligned}\int \frac{1}{(4-x^2)^{\frac{3}{2}}}\,dx &= \int \frac{2\cos\theta}{8\cos^3\theta}\,d\theta \\ &= \frac{1}{4}\int \sec^2\theta\,d\theta \\ &= \frac{1}{4}\tan\theta + C \\ &= \frac{x}{4\sqrt{4-x^2}} + C\end{aligned}$$

Let $x = 2\sin\theta$.

Then $dx = 2\cos\theta\,d\theta$,

and $(4-x^2)^{\frac{3}{2}} = 8\cos^3\theta$.

Here $4 - x^2 = 4\cos^2\theta$. To return to x: $\sin\theta = \frac{x}{2}$ and $\cos\theta = \frac{\sqrt{4-x^2}}{2}$, so $\tan\theta = \frac{x}{\sqrt{4-x^2}}$.

b

$$\begin{aligned}\int_0^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}}\,dx &= \int_{\frac{\pi}{2}}^{\frac{\pi}{6}} \frac{-2\sin\theta}{2\sin\theta}\,d\theta \\ &= \int_{\frac{\pi}{2}}^{\frac{\pi}{6}} (-1)\,d\theta \\ &= \Big[-\theta\Big]_{\frac{\pi}{2}}^{\frac{\pi}{6}} \\ &= -\frac{\pi}{6} + \frac{\pi}{2} \\ &= \frac{\pi}{3}\end{aligned}$$

Let $x = 2\cos\theta$.

Then $dx = -2\sin\theta\,d\theta$,

and $\sqrt{4-x^2} = 2\sin\theta$.

When $x = 0$, $\theta = \frac{\pi}{2}$,

when $x = \sqrt{3}$, $\theta = \frac{\pi}{6}$.

Textbook Homework

1 **c** $\frac{1}{7}(x-1)^7 + \frac{1}{6}(x-1)^6 + C$

2 **a** $\frac{2}{3}(x-1)^{\frac{3}{2}} + 2(x-1)^{\frac{1}{2}} + C$ **b** $\log_e|x-1| - \dfrac{1}{x-1} + C$

3 **c** $\frac{2}{5}(x+1)^{\frac{5}{2}} - \frac{2}{3}(x+1)^{\frac{3}{2}} + C$

4 **a** $\frac{2}{7}(x+1)^{\frac{7}{2}} - \frac{4}{5}(x+1)^{\frac{5}{2}} + \frac{2}{3}(x+1)^{\frac{3}{2}} + C$
b $\frac{4}{3}(x+1)^{\frac{3}{2}} + 2(x+1)^{\frac{1}{2}} + C$

5 **a** $(x+2) - 4\log_e|x+2| + C$ **b** $\frac{1}{3}(2x-1)^{\frac{3}{2}} + 2(2x-1)^{\frac{1}{2}} + C$
c $\frac{3}{40}(4x-5)^{\frac{5}{2}} + \frac{5}{8}(4x-5)^{\frac{3}{2}} + C$ **d** $2(1+\sqrt{x}) - 2\log_e(1+\sqrt{x}) + C$

6 **a** $\frac{49}{20}$ **b** $2\log_e 2 - \frac{1}{2}$ **c** $\frac{8}{9}$ **d** $\frac{1}{9}$ **e** $\frac{128}{15}$ **f** $\frac{4}{3}$ **g** $4 - 6\log_e\frac{5}{3}$ **h** $\frac{2517}{40}$

7 **a** $\sin^{-1}\dfrac{x+2}{3} + C$
b (i) $\dfrac{1}{\sqrt{3}}\tan^{-1}\dfrac{x+1}{\sqrt{3}} + C$ (ii) $\sin^{-1}\dfrac{x+1}{\sqrt{5}} + C$ (iii) $\frac{\pi}{6}$ (iv) $\frac{\pi}{16}$

8 **b** (i) $\frac{1}{3}\tan^{-1}\frac{x}{3}+C$ (ii) $\cos^{-1}\dfrac{x}{\sqrt{3}}+C$ (iii) $\frac{1}{2}\sin^{-1}2x+C$
(iv) $\frac{1}{4}\tan^{-1}4x+C$ (v) $\frac{\pi}{6}$ (vi) $\frac{\pi}{24}$

9 **b** (i) $\dfrac{x}{4\sqrt{4+x^2}}+C$ (ii) $\frac{\pi}{12}-\frac{\sqrt{3}}{8}$ (iii) π
(iv) $-\dfrac{\sqrt{25-x^2}}{25x}+C$ (v) $-\dfrac{\sqrt{9+x^2}}{9x}+C$ (vi) $\frac{\sqrt{3}}{8}$

10 $y=\sqrt{x^2-9}-3\tan^{-1}\dfrac{\sqrt{x^2-9}}{3}$

11 $\frac{1}{3}\left(6\sqrt{3}-7\sqrt{2}\right)$ units^2

13 **b** $\frac{8}{3}$

Integration Part 3 Further Integration - Homework Answers

Lesson 12E | Further integration by substitution

Past-Paper Homework - Worked Solutions

PP 1 **Sydney Girls High School, 2023, Trial Higher School Certificate Examination, Mathematics Extension 1, Q5**

$$\begin{aligned}\int_0^{\sqrt{3}} \frac{1}{\left(x^2+1\right)^2}\,dx &= \int_0^{\frac{\pi}{3}} \frac{\sec^2\theta}{\left(\sec^2\theta\right)^2}\,d\theta \\ &= \int_0^{\frac{\pi}{3}} \frac{1}{\sec^2\theta}\,d\theta \\ &= \int_0^{\frac{\pi}{3}} \cos^2\theta\,d\theta\end{aligned}$$

Let $x=\tan\theta$.

Then $dx=\sec^2\theta\,d\theta$,

and $x^2+1=\sec^2\theta$.

When $x=0$, $\theta=0$,

when $x=\sqrt{3}$, $\theta=\frac{\pi}{3}$.

Answer (A). Here $x^2+1=\tan^2\theta+1=\sec^2\theta$, and $\tan\theta=\sqrt{3}$ gives $\theta=\frac{\pi}{3}$. Option (B), with integrand $\sec^2\theta$, keeps the $\sec^2\theta$ from dx but drops the denominator $\left(\sec^2\theta\right)^2$. Options (C) and (D) use the upper limit $\frac{\pi}{6}$, which belongs to $x=\frac{1}{\sqrt{3}}$, not to $x=\sqrt{3}$.

PP 2 **North Sydney Boys High School, 2022, HSC Course Assessment Task 3-Trial Examination, Mathematics Extension 1, Q4**

Let $y=\sin\theta$, with $0\le\theta\le\frac{\pi}{2}$. Then $dy=\cos\theta\,d\theta$ and $\sqrt{1-y^2}=\sqrt{1-\sin^2\theta}=\cos\theta$. When $y=0$, $\theta=0$, and when $y=\frac{\sqrt{3}}{2}$, $\theta=\frac{\pi}{3}$. So

$$\begin{aligned}\int_0^{\frac{\sqrt{3}}{2}} \sqrt{1-y^2}\,dy &= \int_0^{\frac{\pi}{3}} \cos\theta\times\cos\theta\,d\theta \\ &= \int_0^{\frac{\pi}{3}} \left(\tfrac{1}{2}+\tfrac{1}{2}\cos 2\theta\right)d\theta \\ &= \left[\tfrac{1}{2}\theta+\tfrac{1}{4}\sin 2\theta\right]_0^{\frac{\pi}{3}} \\ &= \left(\tfrac{\pi}{6}+\tfrac{1}{4}\sin\tfrac{2\pi}{3}\right)-\left(0+\tfrac{1}{4}\sin 0\right) \\ &= \tfrac{\pi}{6}+\tfrac{1}{4}\times\tfrac{\sqrt{3}}{2} \\ &= \tfrac{\pi}{6}+\tfrac{\sqrt{3}}{8}\end{aligned}$$

Answer (C). As a check, the integral is the area between the y-axis and the quarter circle $x=\sqrt{1-y^2}$, from $y=0$ to $y=\frac{\sqrt{3}}{2}$. It is a sector of angle $\frac{\pi}{3}$ between the x-axis and the radius to $\left(\frac{1}{2},\frac{\sqrt{3}}{2}\right)$ (area $\frac{1}{2}\times 1^2\times\frac{\pi}{3}=\frac{\pi}{6}$) plus the right-angled triangle with vertices O, $\left(0,\frac{\sqrt{3}}{2}\right)$ and $\left(\frac{1}{2},\frac{\sqrt{3}}{2}\right)$ (area $\frac{1}{2}\times\frac{1}{2}\times\frac{\sqrt{3}}{2}=\frac{\sqrt{3}}{8}$). Option (A) leaves out the triangle. Options (B) and (D) use $\frac{\pi}{3}$, the angle itself, where the primitive $\frac{1}{2}\theta$ gives $\frac{\pi}{6}$.

PP 3 **Girraween High School, 2021, Trial Higher School Certificate Examination, Mathematics Extension 1, Q11(d)**

(i) The square root needs $1-x^2 \geq 0$, and the denominator must not be zero, so $f(x)$ is defined only when $1-x^2 > 0$. It is undefined when $1-x^2 \leq 0$, that is, $x^2 \geq 1$, so $x \leq -1$ or $x \geq 1$.

(ii)

$$\begin{aligned}
\int_0^{\frac{1}{2}} \frac{2x}{\sqrt{1-x^2}}\,dx &= \int_0^{\frac{\pi}{6}} \frac{2\sin u \cos u}{\cos u}\,du \\
&= 2\int_0^{\frac{\pi}{6}} \sin u\,du \\
&= 2\Big[-\cos u\Big]_0^{\frac{\pi}{6}} \\
&= 2\Big(-\cos\tfrac{\pi}{6} + \cos 0\Big) \\
&= 2\Big(-\tfrac{\sqrt{3}}{2} + 1\Big) \\
&= 2-\sqrt{3}
\end{aligned}$$

Let $x = \sin u$.

Then $dx = \cos u\,du$,

and $\sqrt{1-x^2} = \cos u$.

When $x = 0$, $u = 0$,

when $x = \frac{1}{2}$, $u = \frac{\pi}{6}$.

Here $1-x^2 = 1-\sin^2 u = \cos^2 u$, and $\cos u > 0$ for $0 \leq u \leq \frac{\pi}{6}$.

PP 4 **Killara High School, 2024, Year 12 Examination, Mathematics Extension 1 HSC Trial Examination, Q12(b)**

Take $0 \leq \theta \leq \pi$, so that $\sin\theta \geq 0$ and $\sqrt{1-\cos^2\theta} = \sqrt{\sin^2\theta} = \sin\theta$. For the limits, $\theta = \cos^{-1}\left(-\frac{1}{\sqrt{2}}\right) = \frac{3\pi}{4}$ and $\theta = \cos^{-1}\frac{1}{2} = \frac{\pi}{3}$.

Let $x = \cos\theta$. Then $dx = -\sin\theta\, d\theta$, and $\sqrt{1-x^2} = \sin\theta$. When $x = -\frac{1}{\sqrt{2}}$, $\theta = \frac{3\pi}{4}$, when $x = \frac{1}{2}$, $\theta = \frac{\pi}{3}$.

$$\begin{aligned}
\int_{-\frac{1}{\sqrt{2}}}^{\frac{1}{2}} \frac{x}{\sqrt{1-x^2}}\, dx &= \int_{\frac{3\pi}{4}}^{\frac{\pi}{3}} \frac{\cos\theta}{\sin\theta} \times (-\sin\theta)\, d\theta \\
&= -\int_{\frac{3\pi}{4}}^{\frac{\pi}{3}} \cos\theta\, d\theta \\
&= \int_{\frac{\pi}{3}}^{\frac{3\pi}{4}} \cos\theta\, d\theta \\
&= \Big[\sin\theta\Big]_{\frac{\pi}{3}}^{\frac{3\pi}{4}} \\
&= \sin\frac{3\pi}{4} - \sin\frac{\pi}{3} \\
&= \frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2} \\
&= \frac{\sqrt{2}-\sqrt{3}}{2}
\end{aligned}$$

Reversing the limits changes the sign of the integral. The answer is negative because the integrand is negative for $x < 0$, and that part outweighs the positive part.

PP 5 **Caringbah High School, 2020, Higher School Certificate Trial Examination, Mathematics Extension 1, Q14(b)**

(i) Let $x = \cos 2\theta$, with $0 \le \theta \le \frac{\pi}{4}$. Then $dx = -2\sin 2\theta\, d\theta$. When $x = 0$, $\cos 2\theta = 0$, so $2\theta = \frac{\pi}{2}$ and $\theta = \frac{\pi}{4}$; when $x = 1$, $\cos 2\theta = 1$, so $\theta = 0$. Using $1 - \cos 2\theta = 2\sin^2\theta$ and $1 + \cos 2\theta = 2\cos^2\theta$,

$$\sqrt{\frac{1-x}{1+x}} = \sqrt{\frac{2\sin^2\theta}{2\cos^2\theta}} = \frac{\sin\theta}{\cos\theta},$$

because $\sin\theta \ge 0$ and $\cos\theta > 0$ for $0 \le \theta \le \frac{\pi}{4}$. So

$$\begin{aligned}
\int_0^1 \sqrt{\frac{1-x}{1+x}}\, dx &= \int_{\frac{\pi}{4}}^0 \frac{\sin\theta}{\cos\theta} \times (-2\sin 2\theta)\, d\theta \\
&= \int_0^{\frac{\pi}{4}} \frac{\sin\theta}{\cos\theta} \times 2\sin 2\theta\, d\theta \\
&= \int_0^{\frac{\pi}{4}} \frac{\sin\theta}{\cos\theta} \times 4\sin\theta\cos\theta\, d\theta \\
&= \int_0^{\frac{\pi}{4}} 4\sin^2\theta\, d\theta
\end{aligned}$$

The second line reverses the limits and removes the minus sign. The third uses $\sin 2\theta = 2\sin\theta\cos\theta$.

(ii) By part (i), using $\sin^2\theta = \frac{1}{2} - \frac{1}{2}\cos 2\theta$,

$$\begin{aligned}
\int_0^1 \sqrt{\frac{1-x}{1+x}}\, dx &= \int_0^{\frac{\pi}{4}} 4\sin^2\theta\, d\theta \\
&= \int_0^{\frac{\pi}{4}} \left(2 - 2\cos 2\theta\right) d\theta \\
&= \Big[2\theta - \sin 2\theta\Big]_0^{\frac{\pi}{4}} \\
&= \left(\tfrac{\pi}{2} - \sin\tfrac{\pi}{2}\right) - \left(0 - \sin 0\right) \\
&= \tfrac{\pi}{2} - 1
\end{aligned}$$

PP 6 **Abbotsleigh, 2020, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q14(a)**

Take $u \geq 4$. Then $x - 1 = (u-4)^2$, so $\sqrt{x-1} = u - 4$ (the positive square root) and $4 + \sqrt{x-1} = u$. For the limits: when $x = 2$, $(u-4)^2 = 1$, so $u - 4 = 1$ and $u = 5$; when $x = 5$, $(u-4)^2 = 4$, so $u - 4 = 2$ and $u = 6$.

Let $x = (u-4)^2 + 1$. Then $dx = 2(u-4)\,du$, and $4 + \sqrt{x-1} = u$. When $x = 2$, $u = 5$, when $x = 5$, $u = 6$.

$$\begin{aligned}
\int_2^5 \frac{1}{4+\sqrt{x-1}}\,dx &= \int_5^6 \frac{2(u-4)}{u}\,du \\
&= 2\int_5^6 \left(1 - \frac{4}{u}\right)du \\
&= 2\Big[u - 4\log_e u\Big]_5^6 \\
&= 2\big((6 - 4\log_e 6) - (5 - 4\log_e 5)\big) \\
&= 2\big(1 - 4\log_e 6 + 4\log_e 5\big) \\
&= 2 - 8\log_e \tfrac{6}{5}
\end{aligned}$$

The last line uses $\log_e 6 - \log_e 5 = \log_e \frac{6}{5}$. The value is about 0.541.

Integration Part 3 Further Integration - Homework Answers

Lesson 12F | Volumes of rotation

Lesson Examples

C1 **Tutor demo.** Squaring, $y^2 = 25 - x^2$. The limits are $x = 3$ (the line) and $x = 5$ (where the semicircle meets the x-axis).

$$\begin{aligned}
\text{volume} &= \int_3^5 \pi y^2 \, dx \\
&= \pi \int_3^5 \left(25 - x^2\right) dx \\
&= \pi \left[25x - \tfrac{1}{3}x^3\right]_3^5 \\
&= \pi \Big(\left(25 \times 5 - \tfrac{1}{3} \times 5^3\right) - \left(25 \times 3 - \tfrac{1}{3} \times 3^3\right) \Big) \\
&= \pi \left(125 - \tfrac{125}{3} - 75 + 9\right) \\
&= \pi \left(59 - \tfrac{125}{3}\right) \\
&= \tfrac{52\pi}{3} \text{ cubic units}
\end{aligned}$$

Your turn. Squaring, $y^2 = 9 - x^2$. The limits are $x = 1$ and $x = 3$.

$$\begin{aligned}
\text{volume} &= \int_1^3 \pi y^2 \, dx \\
&= \pi \int_1^3 \left(9 - x^2\right) dx \\
&= \pi \left[9x - \tfrac{1}{3}x^3\right]_1^3 \\
&= \pi \Big(\left(9 \times 3 - \tfrac{1}{3} \times 3^3\right) - \left(9 \times 1 - \tfrac{1}{3} \times 1^3\right) \Big) \\
&= \pi \left(27 - 9 - 9 + \tfrac{1}{3}\right) \\
&= \pi \left(9 + \tfrac{1}{3}\right) \\
&= \tfrac{28\pi}{3} \text{ cubic units}
\end{aligned}$$

C2 **Tutor demo.** Because $y = 2x^2$, $x^2 = \frac{1}{2}y$. The limits are $y = 0$ (the vertex) and $y = 8$ (the line).

$$\begin{aligned}\text{volume} &= \int_0^8 \pi x^2 \, dy \\ &= \pi \int_0^8 \tfrac{1}{2} y \, dy \\ &= \pi \left[\tfrac{1}{4} y^2 \right]_0^8 \\ &= \pi \left(\tfrac{1}{4} \times 8^2 - \tfrac{1}{4} \times 0^2 \right) \\ &= 16\pi \text{ cubic units}\end{aligned}$$

Your turn. Because $y = x^2$, $x^2 = y$. The limits are $y = 0$ and $y = 9$.

$$\begin{aligned}\text{volume} &= \int_0^9 \pi x^2 \, dy \\ &= \pi \int_0^9 y \, dy \\ &= \pi \left[\tfrac{1}{2} y^2 \right]_0^9 \\ &= \pi \left(\tfrac{1}{2} \times 9^2 - \tfrac{1}{2} \times 0^2 \right) \\ &= \tfrac{81\pi}{2} \text{ cubic units}\end{aligned}$$

C3 **Tutor demo. a** The parabola is $y_2 = 9 - x^2$. The line AB has gradient $\dfrac{0-9}{3-0} = -3$, so it is $y_1 = 9 - 3x$. For $0 \le x \le 3$, $y_2 - y_1 = 3x - x^2 = x(3-x) \ge 0$. Squaring, $y_2^2 = \left(9 - x^2\right)^2 = 81 - 18x^2 + x^4$ and $y_1^2 = (9 - 3x)^2 = 81 - 54x + 9x^2$.

$$\begin{aligned}\text{volume} &= \int_0^3 \pi \left(y_2^2 - y_1^2 \right) dx \\ &= \pi \int_0^3 \left(\left(81 - 18x^2 + x^4\right) - \left(81 - 54x + 9x^2\right) \right) dx \\ &= \pi \int_0^3 \left(x^4 - 27x^2 + 54x \right) dx \\ &= \pi \left[\tfrac{1}{5} x^5 - 9x^3 + 27x^2 \right]_0^3 \\ &= \pi \left(\left(\tfrac{1}{5} \times 3^5 - 9 \times 3^3 + 27 \times 3^2 \right) - 0 \right) \\ &= \pi \left(\tfrac{243}{5} - 243 + 243 \right) \\ &= \tfrac{243\pi}{5} \text{ cubic units}\end{aligned}$$

b From $y = 9 - x^2$, $x_2^2 = 9 - y$. From $y = 9 - 3x$, $3x = 9 - y$, so $x_1 = 3 - \frac{1}{3}y$ and $x_1^2 = 9 - 2y + \frac{1}{9}y^2$. The limits are $y = 0$ (at B) and $y = 9$ (at A).

$$\begin{aligned}
\text{volume} &= \int_0^9 \pi\left(x_2^2 - x_1^2\right)\,dy \\
&= \pi\int_0^9 \left((9-y) - \left(9 - 2y + \tfrac{1}{9}y^2\right)\right)\,dy \\
&= \pi\int_0^9 \left(y - \tfrac{1}{9}y^2\right)\,dy \\
&= \pi\left[\tfrac{1}{2}y^2 - \tfrac{1}{27}y^3\right]_0^9 \\
&= \pi\left(\left(\tfrac{1}{2}\times 9^2 - \tfrac{1}{27}\times 9^3\right) - 0\right) \\
&= \pi\left(\tfrac{81}{2} - 27\right) \\
&= \tfrac{27\pi}{2}\ \text{cubic units}
\end{aligned}$$

Your turn. a The parabola is $y_2 = 1 - x^2$. The line AB has gradient $\dfrac{0-1}{1-0} = -1$, so it is $y_1 = 1 - x$. For $0 \le x \le 1$, $y_2 - y_1 = x - x^2 = x(1-x) \ge 0$. Squaring, $y_2^2 = \left(1 - x^2\right)^2 = 1 - 2x^2 + x^4$ and $y_1^2 = (1-x)^2 = 1 - 2x + x^2$.

$$\begin{aligned}
\text{volume} &= \int_0^1 \pi\left(y_2^2 - y_1^2\right)\,dx \\
&= \pi\int_0^1 \left(\left(1 - 2x^2 + x^4\right) - \left(1 - 2x + x^2\right)\right)\,dx \\
&= \pi\int_0^1 \left(x^4 - 3x^2 + 2x\right)\,dx \\
&= \pi\left[\tfrac{1}{5}x^5 - x^3 + x^2\right]_0^1 \\
&= \pi\left(\left(\tfrac{1}{5} - 1 + 1\right) - 0\right) \\
&= \tfrac{\pi}{5}\ \text{cubic units}
\end{aligned}$$

b From $y = 1 - x^2$, $x_2^2 = 1 - y$. From $y = 1 - x$, $x_1 = 1 - y$ and $x_1^2 = 1 - 2y + y^2$. The limits are $y = 0$ and $y = 1$.

$$\begin{aligned}\text{volume} &= \int_0^1 \pi\left(x_2^2 - x_1^2\right)\,dy \\ &= \pi\int_0^1 \Big((1-y) - \left(1-2y+y^2\right)\Big)\,dy \\ &= \pi\int_0^1 \left(y - y^2\right)\,dy \\ &= \pi\left[\tfrac{1}{2}y^2 - \tfrac{1}{3}y^3\right]_0^1 \\ &= \pi\Big(\left(\tfrac{1}{2} - \tfrac{1}{3}\right) - 0\Big) \\ &= \tfrac{\pi}{6} \text{ cubic units}\end{aligned}$$

C4 **Tutor demo.** At $x = 6$, $y = \frac{1}{2} \times 6 = 3$. So $r = 3$ and $h = 6$.
Squaring, $y^2 = \frac{1}{4}x^2$.

$$\begin{aligned}\text{volume} &= \int_0^6 \pi y^2\,dx \\ &= \pi\int_0^6 \tfrac{1}{4}x^2\,dx \\ &= \pi\left[\tfrac{1}{12}x^3\right]_0^6 \\ &= \pi\left(\tfrac{1}{12} \times 6^3 - 0\right) \\ &= 18\pi \text{ cubic units}\end{aligned}$$

Check: $\frac{1}{3}\pi r^2 h = \frac{1}{3} \times \pi \times 3^2 \times 6 = 18\pi$.
Your turn. The semicircle meets the x-axis at $x = \pm 4$, so the radius is $r = 4$.
Squaring, $y^2 = 16 - x^2$.

$$\begin{aligned}\text{volume} &= \int_{-4}^4 \pi y^2\,dx \\ &= \pi\int_{-4}^4 \left(16 - x^2\right)\,dx \\ &= \pi\left[16x - \tfrac{1}{3}x^3\right]_{-4}^4 \\ &= \pi\Big(\left(64 - \tfrac{64}{3}\right) - \left(-64 + \tfrac{64}{3}\right)\Big) \\ &= \pi\left(128 - \tfrac{128}{3}\right) \\ &= \tfrac{256\pi}{3} \text{ cubic units}\end{aligned}$$

Check: $\frac{4}{3}\pi r^3 = \frac{4}{3} \times \pi \times 4^3 = \frac{256\pi}{3}$.

Textbook Homework

1 **b** radius 9, height 3, volume 81π u^3 **c** 81π

2 **b** radius 3, volume 36π u^3 **c** 36π

3 **a** 16π u^3 **b** 9π u^3 **c** $\frac{32\pi}{5}$ u^3 **d** 6π u^3 **e** $\frac{16\pi}{3}$ u^3 **f** $\frac{\pi}{7}$ u^3 **g** 9π u^3 **h** 16π u^3

4 **a** 3π u^3 **b** $\frac{256\pi}{3}$ u^3 **c** $\frac{3093\pi}{5}$ u^3 **d** $\frac{\pi}{2}$ u^3 **e** $\frac{256\pi}{3}$ u^3 **f** $\frac{243\pi}{5}$ u^3 **g** $\frac{16\pi}{15}$ u^3 **h** $\frac{16\pi}{3}$ u^3

5 $\frac{\pi}{2}\left(e^2-1\right)$ cubic units

6 $\pi \ln 2$ u^3

7 $\pi \ln 6$ u^3

8 **a** $\tan^2 x = \sec^2 x - 1$ **b** $\pi\left(\sqrt{3}-\frac{\pi}{3}\right)$ u^3

9 **a** $\sin^2 x = \frac{1}{2} - \frac{1}{2}\cos 2x$ **b** $\frac{\pi^2}{4}$ u^3

10 **a** $\frac{296\pi}{3}$ u^3 **b** $\frac{119\pi}{6}$ u^3 **c** $\frac{625\pi}{6}$ u^3 **d** $\frac{16\pi}{105}$ u^3

11 **a** $\frac{\pi}{3}$ u^3 **b** $\frac{28\pi}{15}$ u^3 **c** $\frac{81\pi}{10}$ u^3 **d** $\frac{\pi}{2}$ u^3

14 $16\pi(3\pi-8) \doteq 71.62$ mL

15 **a** 256π u^3 **b** 128π u^3 **c** 128π u^3

16 **a** $\frac{32\pi}{5}$ u^3, 8π u^3 **b** $\frac{50\pi}{3}$ u^3, $\frac{5\pi}{3}$ u^3 **c** 8π u^3, $\frac{128\pi}{5}$ u^3 **d** $\frac{24\pi}{5}$ u^3, $\frac{\pi}{2}$ u^3

17 **a** The curves meet at $(0,0)$ and $(1,1)$, and $y=x^2$ is above $y=x^3$ for $0<x<1$.
b (i) $\frac{2\pi}{35}$ u^3 (ii) $\frac{\pi}{10}$ u^3

18 **a** The graphs meet at $(1,5)$ and $(5,1)$. **b** $\frac{64\pi}{3}$ u^3

19 $\pi(8\ln 2 - 5)$ u^3

20 **a** $y=3x$ **b** The region lies between $x=0$ and $x=1$, with the curve above the tangent; they touch at $(1,3)$. **c** (i) $\frac{15\pi}{7}$ u^3 (ii) $\frac{2\pi}{5}$ u^3

21 $\frac{\pi}{2}(8\ln 2 - 3)$ u^3

23 **a** $\ln(\sec\theta+\tan\theta)+C$

24 **a** $(0,0)$ and $(1,1)$
c It is the cone formed by rotating the line $y=x$ from $x=0$ to $x=1$ about the x-axis. **d** $\frac{\pi}{3}$ u^3

25 **a** The parabola meets the x-axis at $x=2$ and $x=4$, with vertex $(3,1)$. **c** 8π u^3

Integration Part 3 Further Integration - Homework Answers

Lesson 12F | Volumes of rotation

Past-Paper Homework - Worked Solutions

PP 1 **Sydney Boys High School, 2020, Year 12 Term 3 Trial HSC, Mathematics Extension 1, Q8**

The rotation is about the y-axis, so the volume is $\pi \int_1^2 x^2\,dy$, and x^2 must be written in terms of y:

$$y = e^{\frac{x}{2}} \quad \Rightarrow \quad \frac{x}{2} = \ln y \quad \Rightarrow \quad x = 2\ln y \quad \Rightarrow \quad x^2 = 4(\ln y)^2.$$

$$\text{So} \quad V = \pi \int_1^2 4(\ln y)^2\,dy = 4\pi \int_1^2 (\ln y)^2\,dy.$$

Options A and B have $4y\ln y$ and $4\ln y^2 = 8\ln y$ in place of $4(\ln y)^2$. Option C integrates in x with limits that are values of y. **Answer D.**

PP 2 **Caringbah High School, 2021, Year 12 Mathematics Extension 1 HSC Course Assessment Task 4, Q12(a)**

Squaring, $y^2 = \dfrac{4}{x-7}$. So

$$\begin{aligned} V &= \pi \int_8^{10} y^2\,dx \\ &= \pi \int_8^{10} \frac{4}{x-7}\,dx \\ &= 4\pi \Big[\ln(x-7)\Big]_8^{10} \\ &= 4\pi\big(\ln(10-7) - \ln(8-7)\big) \\ &= 4\pi(\ln 3 - \ln 1) \\ &= 4\pi \ln 3 \text{ cubic units} \end{aligned}$$

On $8 \le x \le 10$, $x - 7 > 0$, so no absolute value is needed, and $\ln 1 = 0$.

PP 3 **North Sydney Boys High School, 2020, Year 12 Assessment Task 3, Mathematics Extension 1, Q8**

Rotation about the y-axis: write each curve with x^2 as the subject. On the shaded region $x \geq 1$, so for the parabola take the positive square root:

$$\begin{aligned} y = (x-1)^2 &\Rightarrow x - 1 = \sqrt{y} \Rightarrow x = \sqrt{y} + 1 \\ &\Rightarrow x^2 = \left(\sqrt{y}+1\right)^2 = y + 2\sqrt{y} + 1, \\ y = \log_e x &\Rightarrow x = e^y \Rightarrow x^2 = e^{2y}. \end{aligned}$$

The region runs from $y = 0$ (at B) to $y = \ln a$ (at A). Across it the parabola is further from the y-axis (for example at $y = 0.25$: $\sqrt{0.25} + 1 = 1.5$ and $e^{0.25} \doteq 1.28$), so $x_2 = \sqrt{y} + 1$ and $x_1 = e^y$ in Box 13:

$$V = \pi \int_0^{\ln a} \left(x_2^2 - x_1^2\right) dy = \pi \int_0^{\ln a} \left(y + 2\sqrt{y} + 1 - e^{2y}\right) dy.$$

Option A squares the difference of the radii, which is not the difference of the squares. Option C is a formula for rotation about the x-axis. Option D does not square the radii.

Answer B.

PP 4 **Killara High School, 2024, Year 12 Examination, Mathematics Extension 1 HSC Trial Examination, Q13(a)**

The curve is above the x-axis on $\left[0, \frac{\pi}{2}\right]$ and meets it at both ends. Square $f(x)$, using $\sin 2x = 2\sin x \cos x$ and then $\sin^2\theta = \frac{1}{2} - \frac{1}{2}\cos 2\theta$ with $\theta = 2x$:

$$\begin{aligned} \left(f(x)\right)^2 &= \sin^2 x \cos^2 x \\ &= \left(\tfrac{1}{2}\sin 2x\right)^2 \\ &= \tfrac{1}{4}\sin^2 2x \\ &= \tfrac{1}{4}\left(\tfrac{1}{2} - \tfrac{1}{2}\cos 4x\right) \\ &= \tfrac{1}{8}(1 - \cos 4x). \end{aligned}$$

$$\begin{aligned} V &= \pi \int_0^{\frac{\pi}{2}} \left(f(x)\right)^2 dx \\ &= \frac{\pi}{8} \int_0^{\frac{\pi}{2}} (1 - \cos 4x)\, dx \\ &= \frac{\pi}{8} \left[x - \tfrac{1}{4}\sin 4x\right]_0^{\frac{\pi}{2}} \\ &= \frac{\pi}{8} \left(\left(\tfrac{\pi}{2} - \tfrac{1}{4}\sin 2\pi\right) - \left(0 - \tfrac{1}{4}\sin 0\right)\right) \\ &= \frac{\pi}{8} \times \frac{\pi}{2} \\ &= \frac{\pi^2}{16} \text{ cubic units} \end{aligned}$$

PP 5 **Blacktown Boys' High School, 2020, HSC Trial Examination, Mathematics Extension 1, Q14(b)**

(i) By the chain rule, with $u = 2\sin x$ and $\dfrac{du}{dx} = 2\cos x$,

$$\frac{d}{dx}\left(e^{2\sin x}\right) = e^{2\sin x} \times 2\cos x = 2\cos x\, e^{2\sin x}.$$

(ii) Squaring, $y^2 = \left(\sqrt{\cos x}\right)^2 \left(e^{\sin x}\right)^2 = \cos x\, e^{2\sin x}$. Using part (i), a primitive of $2\cos x\, e^{2\sin x}$ is $e^{2\sin x}$:

$$\begin{aligned}
V &= \pi \int_0^{\frac{\pi}{2}} y^2\, dx \\
&= \pi \int_0^{\frac{\pi}{2}} \cos x\, e^{2\sin x}\, dx \\
&= \frac{\pi}{2} \int_0^{\frac{\pi}{2}} 2\cos x\, e^{2\sin x}\, dx \\
&= \frac{\pi}{2} \left[e^{2\sin x}\right]_0^{\frac{\pi}{2}} \\
&= \frac{\pi}{2} \left(e^{2\sin\frac{\pi}{2}} - e^{2\sin 0}\right) \\
&= \frac{\pi}{2} \left(e^2 - e^0\right) \\
&= \frac{\pi}{2} \left(e^2 - 1\right) \text{ cubic units}
\end{aligned}$$

PP 6 **Sydney Technical High School, 2021, Extension 1 Mathematics, Trial HSC, Q13(c)**

The region is bounded on the right by $y = x^2$ for $0 \le y \le 1$, and by $y = 5 - 4x^2$ for $1 \le y \le 5$ (they meet at $(1, 1)$, and $y = 5 - 4x^2$ meets the y-axis at 5). Write x^2 as the subject:

$$y = x^2 \;\Rightarrow\; x^2 = y, \qquad y = 5 - 4x^2 \;\Rightarrow\; 4x^2 = 5 - y \;\Rightarrow\; x^2 = \frac{5 - y}{4}.$$

$$\begin{aligned}
V &= \pi \int_0^1 y\, dy + \pi \int_1^5 \frac{5 - y}{4}\, dy \\
&= \pi \left[\tfrac{1}{2} y^2\right]_0^1 + \frac{\pi}{4} \left[5y - \tfrac{1}{2} y^2\right]_1^5 \\
&= \pi\left(\tfrac{1}{2} - 0\right) + \frac{\pi}{4}\left(\left(25 - \tfrac{25}{2}\right) - \left(5 - \tfrac{1}{2}\right)\right) \\
&= \frac{\pi}{2} + \frac{\pi}{4}\left(\tfrac{25}{2} - \tfrac{9}{2}\right) \\
&= \frac{\pi}{2} + \frac{\pi}{4} \times 8 \\
&= \frac{\pi}{2} + 2\pi \\
&= \frac{5\pi}{2} \text{ cubic units}
\end{aligned}$$

PP 7 **Sydney Girls High School, 2023, Trial Higher School Certificate Examination, Mathematics Extension 1, Q12(e)**

The line and the parabola meet where $x^2 = \frac{1}{2}x + 3$, that is $2x^2 - x - 6 = 0$, $(2x+3)(x-2) = 0$. In the first quadrant $x = 2$, so they meet at $(2, 4)$. The line meets the y-axis at $(0, 3)$. The region lies between the parabola (below) and the line (above), for $0 \leq x \leq 2$.

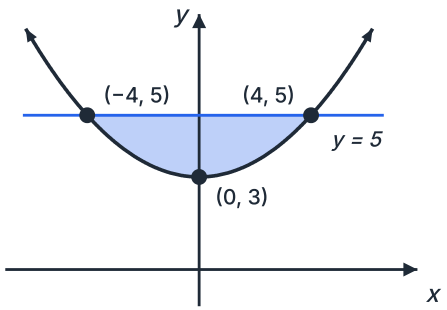
As functions of y: the parabola is $x = \sqrt{y}$, so $x^2 = y$; the line is $x = 2y - 6$. For $0 \leq y \leq 3$, the region runs from the y-axis to the parabola. For $3 \leq y \leq 4$, it runs from the line $x_1 = 2y - 6$ to the parabola $x_2 = \sqrt{y}$. Also $(2y-6)^2 = 4y^2 - 24y + 36$.

$$
\begin{aligned}
V &= \pi \int_0^3 y\,dy + \pi \int_3^4 \left(y - (2y-6)^2\right) dy \\
&= \pi\left[\tfrac{1}{2}y^2\right]_0^3 + \pi \int_3^4 \left(-4y^2 + 25y - 36\right) dy \\
&= \frac{9\pi}{2} + \pi\left[-\tfrac{4}{3}y^3 + \tfrac{25}{2}y^2 - 36y\right]_3^4 \\
&= \frac{9\pi}{2} + \pi\left(\left(-\tfrac{256}{3} + 200 - 144\right) - \left(-36 + \tfrac{225}{2} - 108\right)\right) \\
&= \frac{9\pi}{2} + \pi\left(-\tfrac{88}{3} + \tfrac{63}{2}\right) \\
&= \frac{9\pi}{2} + \frac{13\pi}{6} \\
&= \frac{20\pi}{3} \text{ cubic units}
\end{aligned}
$$

$\pi\left[\frac{1}{2}y^2\right]_0^3 = \pi\left(\frac{9}{2} - 0\right)$. At $y = 4$: $-\frac{4}{3} \times 64 + \frac{25}{2} \times 16 - 36 \times 4$. At $y = 3$: $-\frac{4}{3} \times 27 + \frac{25}{2} \times 9 - 36 \times 3$.

PP 8 **James Ruse Agricultural High School, 2020, Year 12 Trial HSC Examination, Mathematics Extension 1, Q12(a)**

(i) $x^2 = 8(y-3)$ gives $y = 3 + \frac{1}{8}x^2$: a concave-up parabola with vertex $(0, 3)$ and axis the y-axis.



(ii) The line $y = 5$ meets the parabola where $x^2 = 8(5-3) = 16$, so at $(-4, 5)$ and $(4, 5)$. The region lies between $y_2 = 5$ (above) and $y_1 = 3 + \frac{1}{8}x^2$ (below) for $-4 \le x \le 4$, and both are positive, so Box 13 applies.

Squaring, $y_1^2 = \left(3 + \frac{1}{8}x^2\right)^2 = 9 + \frac{3}{4}x^2 + \frac{1}{64}x^4$, and $y_2^2 = 25$. The integrand is even, so

$$
\begin{aligned}
V &= \pi \int_{-4}^{4} \left(y_2^2 - y_1^2\right) dx \\
&= \pi \int_{-4}^{4} \left(16 - \tfrac{3}{4}x^2 - \tfrac{1}{64}x^4\right) dx \\
&= 2\pi \int_{0}^{4} \left(16 - \tfrac{3}{4}x^2 - \tfrac{1}{64}x^4\right) dx \\
&= 2\pi \left[16x - \tfrac{1}{4}x^3 - \tfrac{1}{320}x^5\right]_0^4 \\
&= 2\pi \left(\left(64 - 16 - \tfrac{1024}{320}\right) - 0\right) \\
&= 2\pi \left(48 - \tfrac{16}{5}\right) \\
&= \frac{448\pi}{5} \text{ cubic units}
\end{aligned}
$$

The same answer comes from the cylinder of radius 5 and length 8 (200π) minus the solid under the parabola: $200\pi - 2\pi\left(\frac{16}{5} + 16 + 36\right) = \frac{448\pi}{5}$.

Integration Part 3 Further Integration - Homework Answers

Review | Chapter review exercise

1 **a** $\dfrac{3}{\sqrt{1-9x^2}}$ **b** $\dfrac{3}{9+x^2}$ **c** $\dfrac{1}{\sqrt{2x-x^2}}$ **d** $2x\tan^{-1}x+\dfrac{x^2}{1+x^2}$
e $\dfrac{2}{x^2+4x+8}$ **f** $\dfrac{-1}{\sqrt{x^4-x^2}}$

4 **a** $\dfrac{1}{\sqrt{3}}\tan^{-1}\dfrac{x}{\sqrt{3}}+C$ **b** $\sin^{-1}\dfrac{x}{\sqrt{3}}+C$ **c** $\dfrac{1}{6}\tan^{-1}\dfrac{2x}{3}+C$
d $\dfrac{1}{3}\cos^{-1}\dfrac{3x}{4}+C$

5 **a** $\frac{\pi}{36}$ **b** $\frac{\pi}{3}$

6 **a** $\frac{1}{2}x+\frac{1}{4}\sin 2x+C$ **b** $\frac{1}{2}x-\frac{1}{4}\sin 2x+C$ **c** $\frac{1}{2}x+\frac{1}{8}\sin 4x+C$
d $\frac{1}{2}x-\frac{1}{16}\sin 8x+C$

7 **a** $\frac{\pi}{6}$ **b** $\frac{1}{12}(\pi+3)$

9 **a** $\frac{1}{6}(5x-1)^6+C$ **b** $\frac{1}{3}\left(x^2+2\right)^3+C$ **c** $\dfrac{-1}{x^4+1}+C$ **d** $\frac{1}{2}\sqrt{4x+3}+C$
e $\frac{1}{3}\sin^3 x+C$ **f** $\frac{1}{4}\tan^4 x+C$

10 **a** $\frac{1}{15}$ **b** $\frac{1}{4}$ **c** $\frac{1}{3}$ **d** $\frac{1}{3}$ **e** e^2-e **f** $\frac{1}{2}\ln 2$

11 **a** $x-1+\ln|x-1|+C$ **b** $\frac{2}{3}(x+2)^{\frac{3}{2}}-6(x+2)^{\frac{1}{2}}+C$
c $\frac{1}{10}(2x+1)^{\frac{5}{2}}-\frac{1}{6}(2x+1)^{\frac{3}{2}}+C$ **d** $2\left(4+\sqrt{x}\right)-8\ln\left(4+\sqrt{x}\right)+C$

12 **a** $\frac{11}{30}$ **b** $4-3\ln 2$ **c** 36 **d** $\frac{13}{15}$

13 **a** domain $x\leq 9$, range $y\geq 0$ **b** The upper half of a parabola on its side, from $(9,0)$ through $(0,3)$. **c** 18 u^2 **d** (i) $\frac{81\pi}{2}$ u^3 (ii) $\frac{648\pi}{5}$ u^3

14 $4\pi\ln 2$ u^3

15 $\frac{2688\pi}{5}$ u^3

16 **a** $\cos^2 2x=\frac{1}{2}+\frac{1}{2}\cos 4x$

17 8.49 u^3

18 **a** R has vertices $(0,0)$, $(0,1)$ and $\left(\frac{\pi}{4},0\right)$, with the curve as its third side.
b $\pi(1-\ln 2)$ u^3

19 **a** Minimum turning point at $(1,2)$, maximum turning point at $(-1,-2)$. **c** $\frac{9\pi}{4}$ u^3

21 **a** $\pi\displaystyle\int_1^3 2^{2x+2}\,dx$

b 180π u^3

c $\dfrac{120\pi}{\log_e 2} \doteq 173\pi$ u^3. The curve is concave up, so each trapezium lies above the curve and overestimates the area.